# Acknowledgment of **Country**

The University of Queensland (UQ) acknowledges the Traditional Owners and their custodianship of the lands on which we meet.

We pay our respects to their Ancestors and their descendants, who continue cultural and spiritual connections to Country.

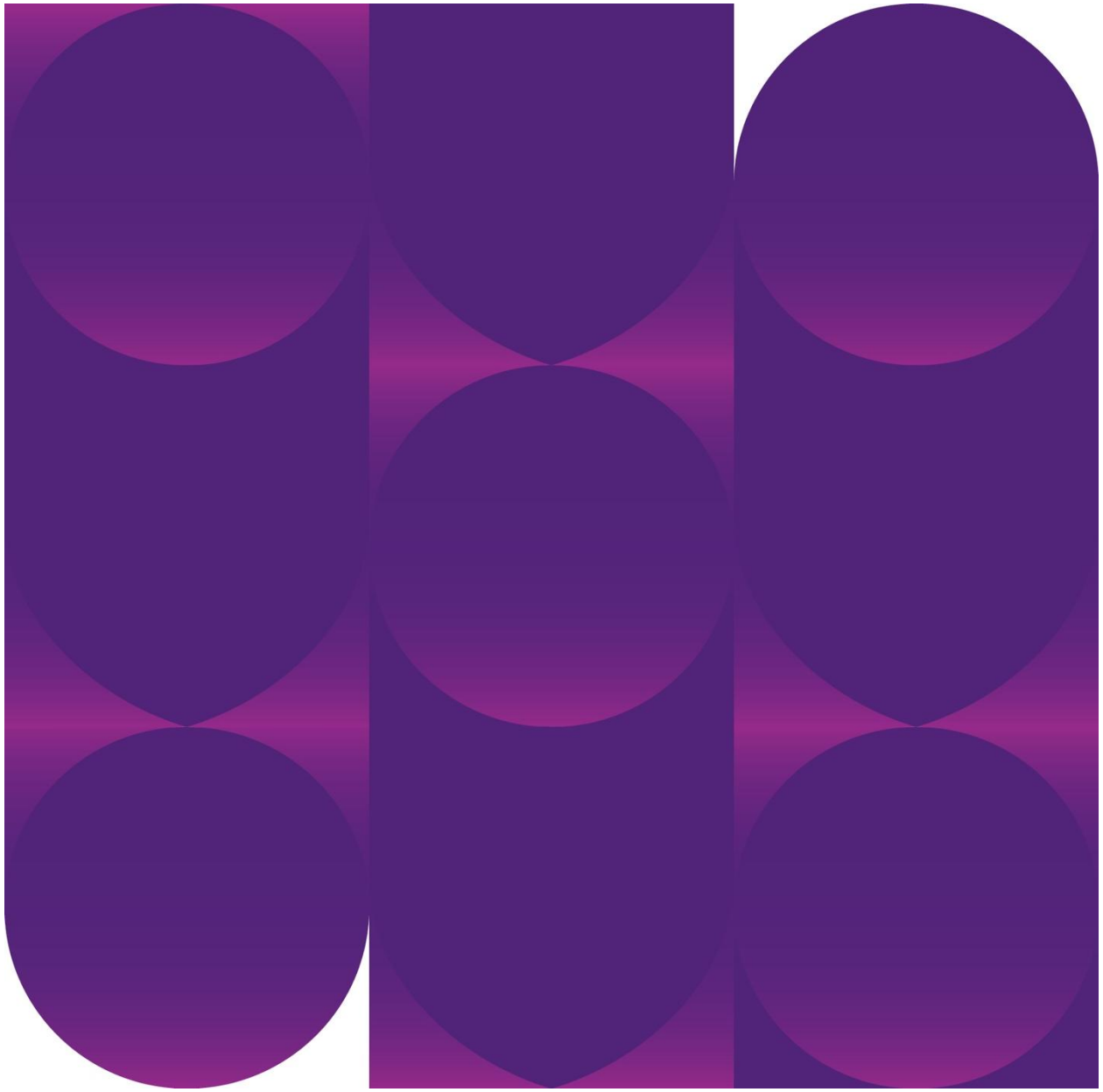
We recognise their valuable contributions to Australian and global society.



THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

# Nanocellulose

**From plant fiber to functional, sustainable material**



MATE6301 · Nanomaterials

A/Prof Nasim Amiralian

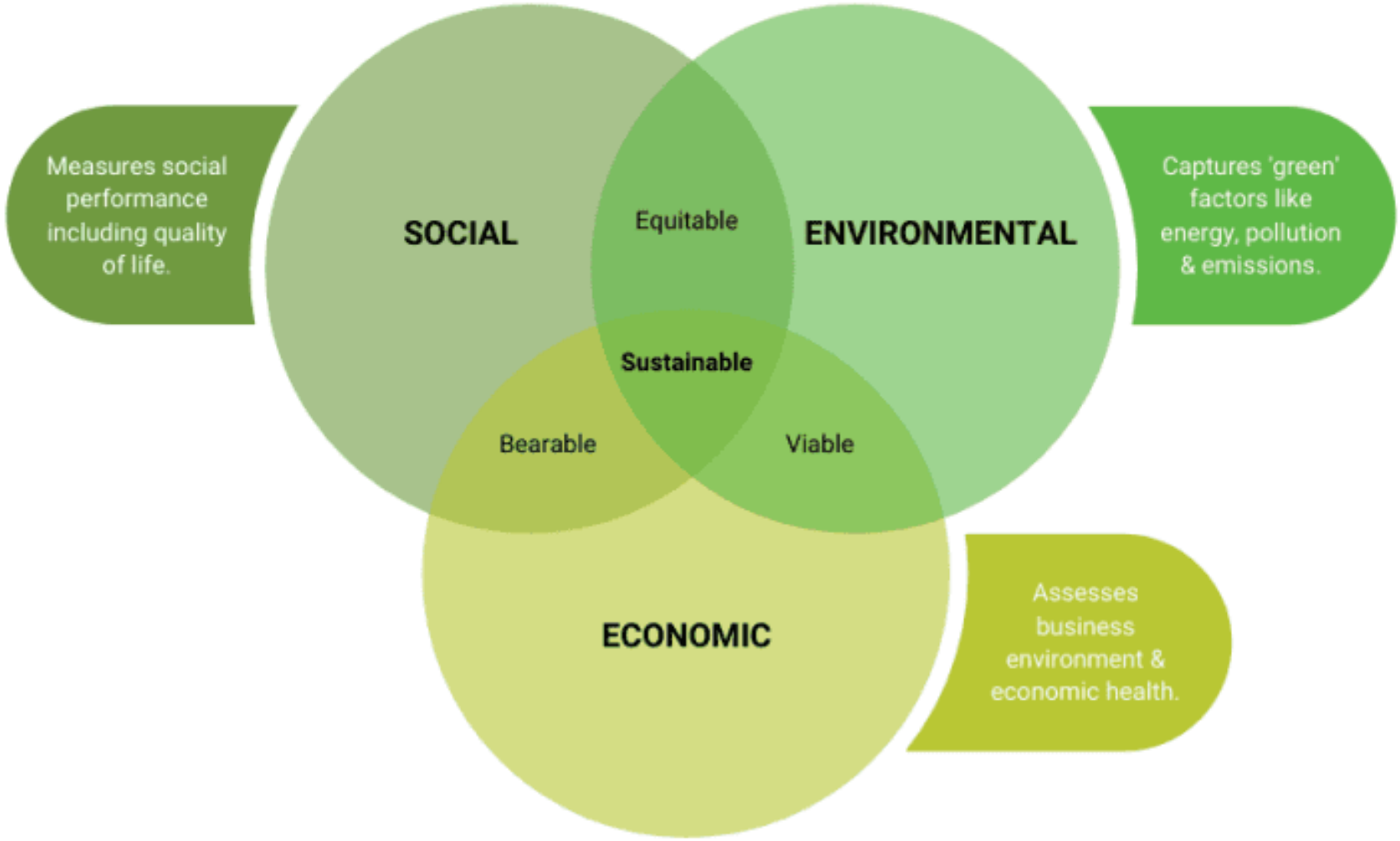
# What we'll cover today

- Why materials science needs an alternative to fossil-based plastics
- What cellulose is, and the different natural sources it comes from
- How nanocellulose is made — from plant fibre to nanoscale material
- How we characterise nanocellulose, and the challenges it still faces
- Where nanocellulose is already being used, commercially and in research
- Two UQ case studies: spinifex nanocellulose and antimicrobial cellulose coatings

# Sustainability

Sustainability means **meeting our own needs without compromising the ability of future generations to meet their own needs**.






https://www.mcgill.ca/sustainability/files/sustainability/what-is-sustainability.pdf

https://www.sustain.ucla.edu/what-is-sustainability/

# Excessive consumption



Climate Change
Greenhouse Gas Emissions



Plastic pollution
Microplastics




Economic Costs



Environmental pollution
Resource depletion



Social impact
Declining ecosystem health

# Impact of plastic pollution: key statistics

**Global Production**: Over 8.3 billion metric tons of plastic have been produced since the 1950s.

**Recycling Rates**: Only about 9% of plastic waste has been recycled globally.

**Landfills and Environment**: Approximately 79% of plastic waste accumulates in landfills or the natural environment.

**Incineration**: About 12% of plastic waste is incinerated.

**Economic Cost**: Plastic pollution costs the global economy an estimated $13 billion annually.

**Microplastics**: Microplastics are found in 83% of tap water samples worldwide.

**Air Pollution**: Incineration of plastics releases harmful chemicals like dioxins, furans, and PCBs.

**Human Health**: Over 90% of humans have plastic particles in their bodies.

**Wildlife Impact**: 100,000 marine mammals and 1 million seabirds die annually due to plastic pollution.

**Soil Contamination**: Up to 23% of plastic waste can end up in agricultural soils.

**Legislation**: More than 60 countries have implemented bans or levies on single-use plastics.

https://ourworldindata.org/plastic-pollution | *Nature* 619, 222 (2023) | https://www.oecd-ilibrary.org/environment/data/global-plastic-outlook_c0821f81-en

# Did you know?

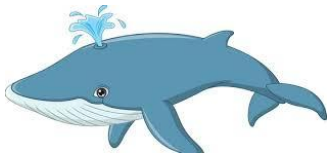
Single use plastic production per year = 300 million tons

=

Entire human population

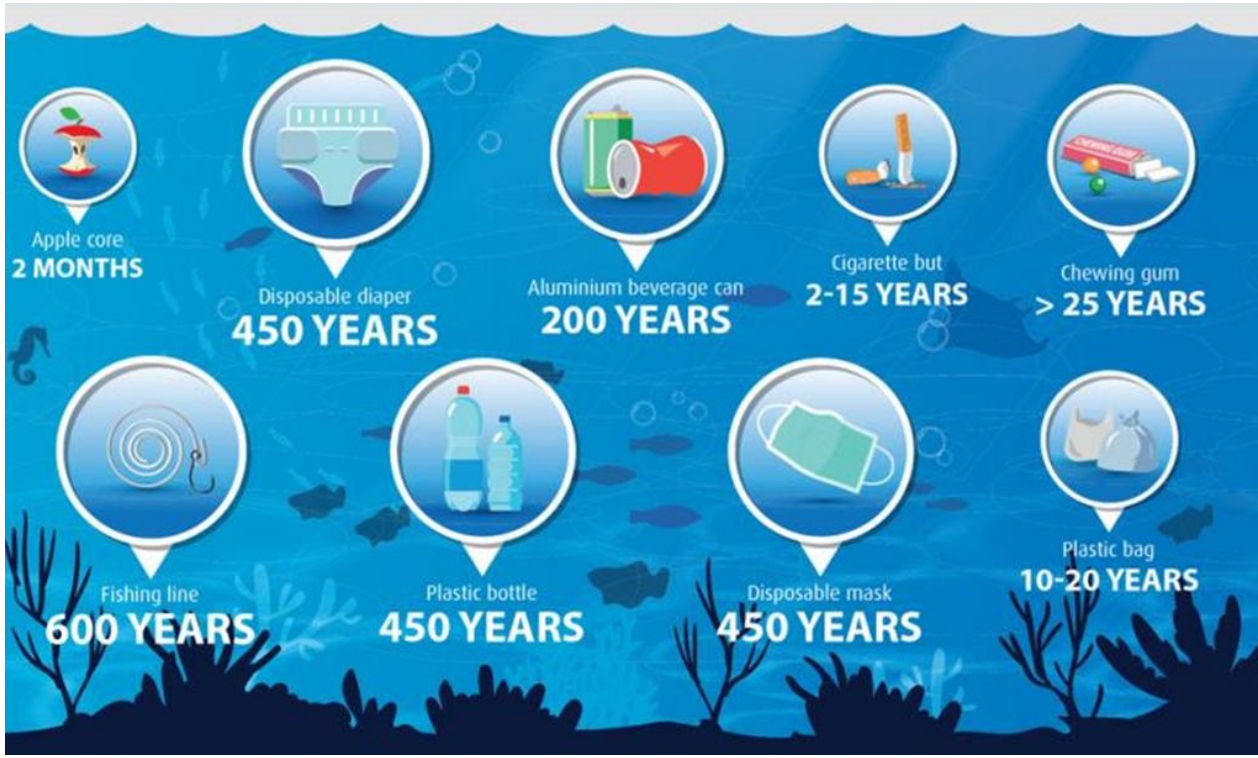


2,700,000 blue whales



681 million football pitches






**Did you know?**

10M tons of plastic are dumped in the ocean/pers = more than a garbage truck every minute!

**Did you know?**

There are more particles of microplastics in the ocean than the stars in the milky way.

**Did you know?**

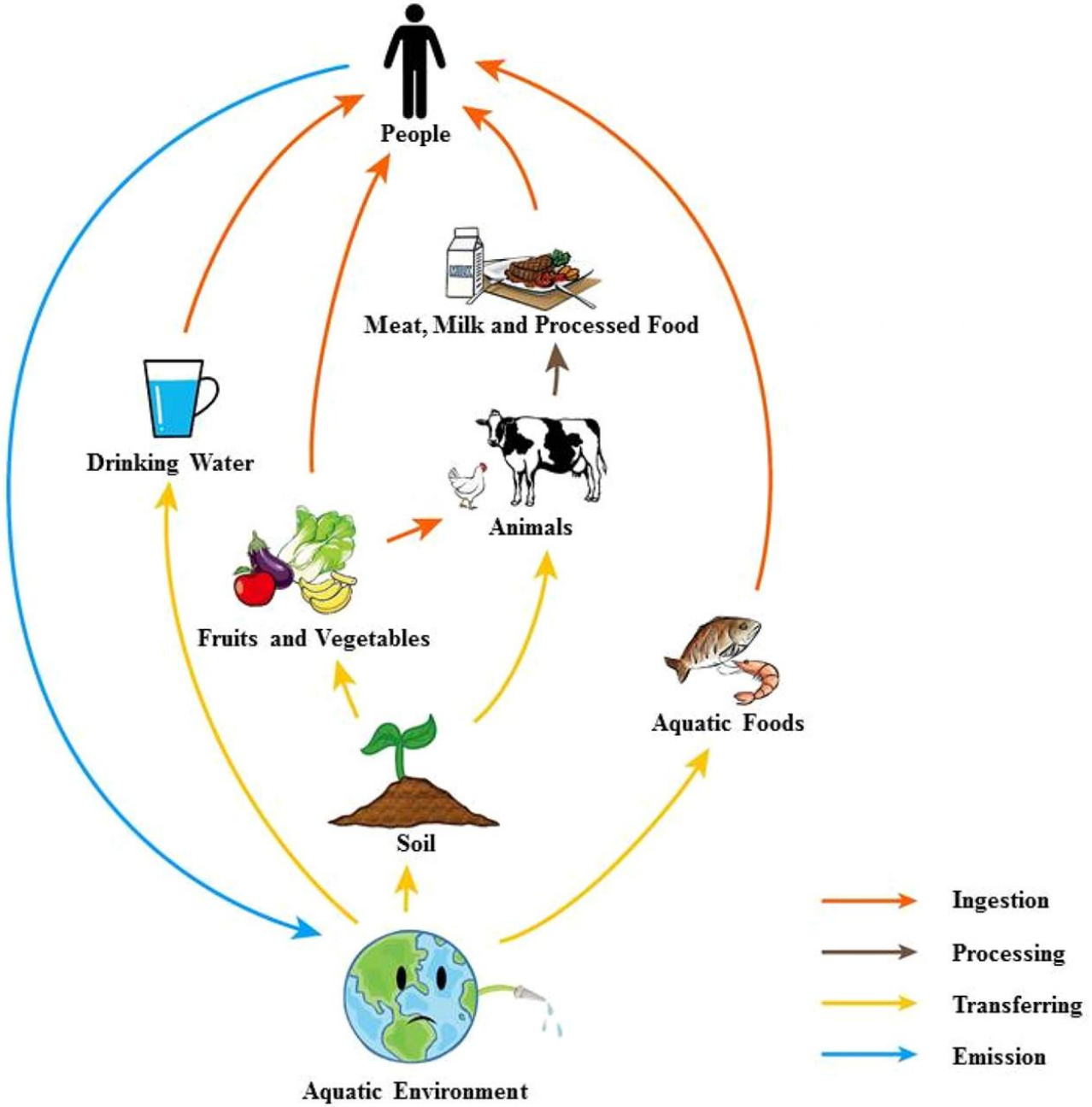
By 2050, there will be more plastic in the ocean than fish.

**Did you know?**

100% of mussels tested have microplastics!

# Microplastics in the food chain

Humans eat over 18 kg of plastic in their lifetime.



# The problem: linear, fossil-based materials

**Fossil feedstock**

Oil and gas are the primary carbon input for most plastics, coatings and packaging

→

**Manufacture**

High-throughput production optimised for cost and performance, not end-of-life

→

**Short use phase**

A large share of plastic is single-use, discarded within months or even minutes

→

**Waste & pollution**

Recycling alone cannot solve design, contamination and collection limits

**Nanocellulose asks: can a renewable, nanostructured material deliver equivalent function with a lower lifecycle burden?**

# Cellulose

Cellulose is the building block of plants.

The most abundant polymer is available worldwide($10^{12}$ tons produced per year).

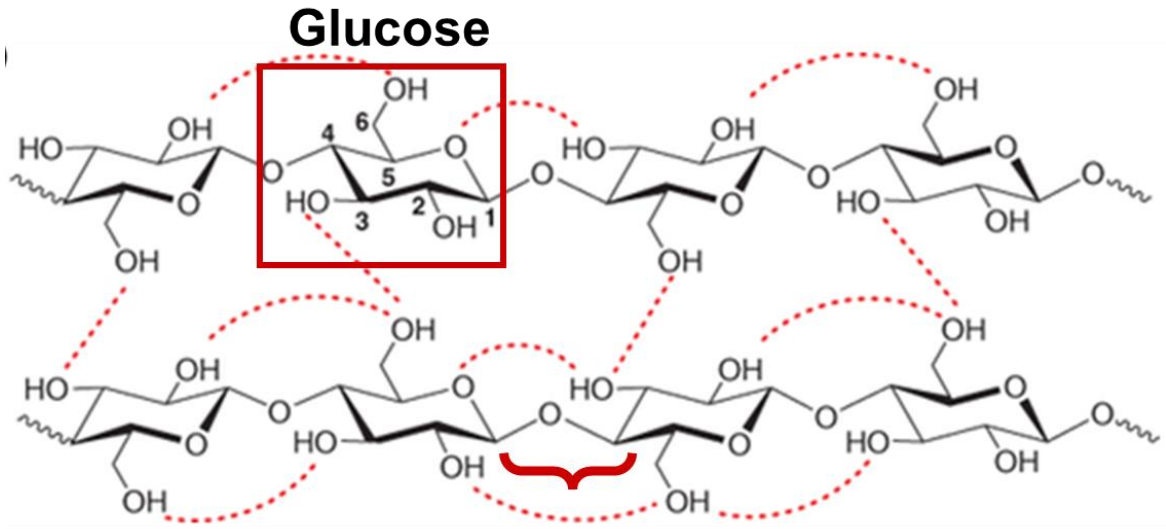
Cellulose is a polysaccharide made up of thousands of D-glucose subunits linked via beta 1-4 glycosidic bonds.

Linear homopolymer

Forms semi-crystalline microfibrils

Insoluble





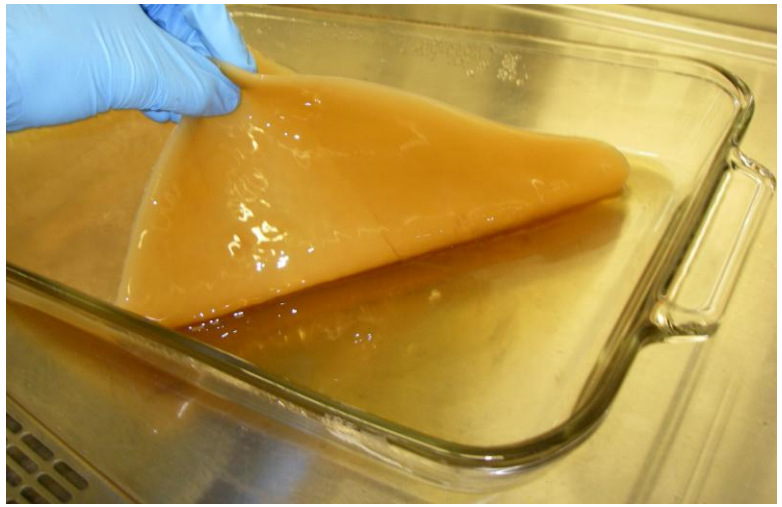

1-4 beta glycosidic bonds

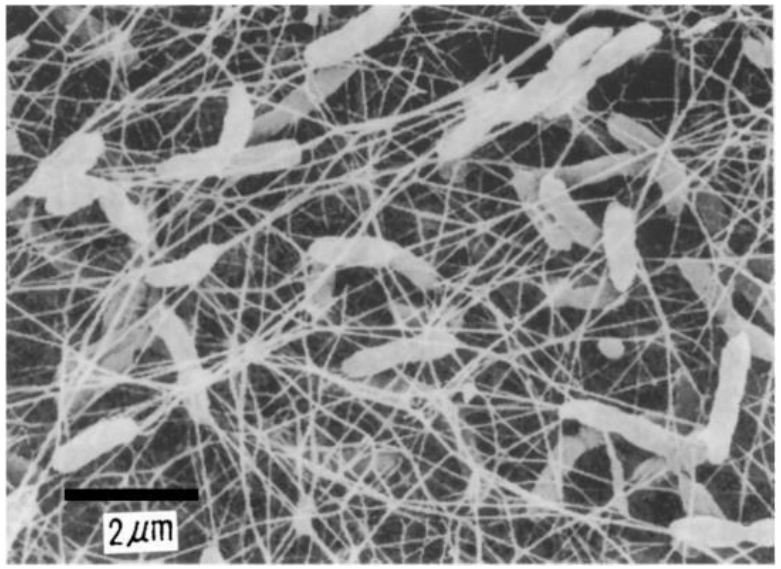
# Source of cellulose




Industrial Crops and Products, 93(25),2016, 26-38

# Bacterial cellulose

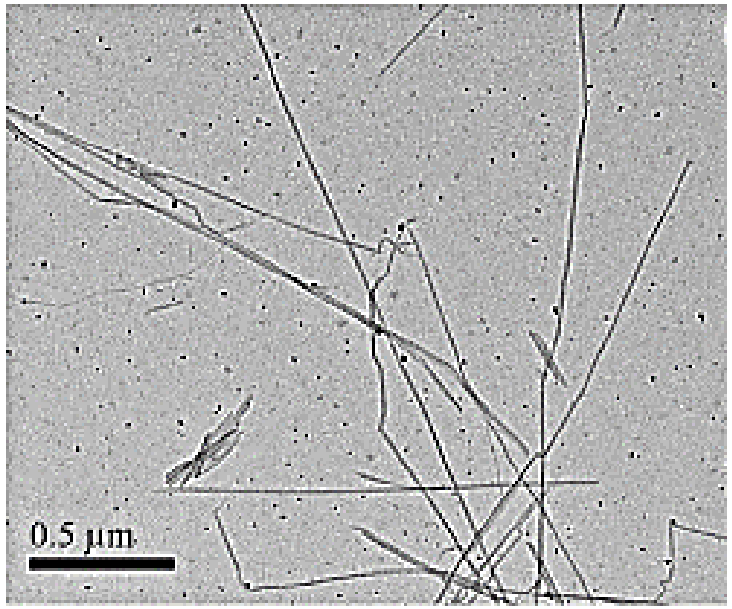





- Some species of bacteria can biosynthesise cellulose (notably *acetobacteria xylinum*)
- The cellulose is extracellular, it is not in the bacteria but extruded by the bacteria
- Bacterial cellulose is the only pure form of cellulose in nature, no other components are synthesised with cellulose
- Microfibrils in bacterial cellulose are generally larger than in other cellulose grades

# Tunicate cellulose






Tunicates are small marine animals

- They are the *only species of animals* that biosynthesize cellulose

- The body of a tunicate is surrounded by a test or a tunic where the cellulose is Produced

- Tunicate cellulose is in the form of microfibrils that are highly crystalline compared with most plant celluloses

# Plant cellulose

**Herbaceous plants**

- Agro fibres
- At least the part above ground dies after the growing season

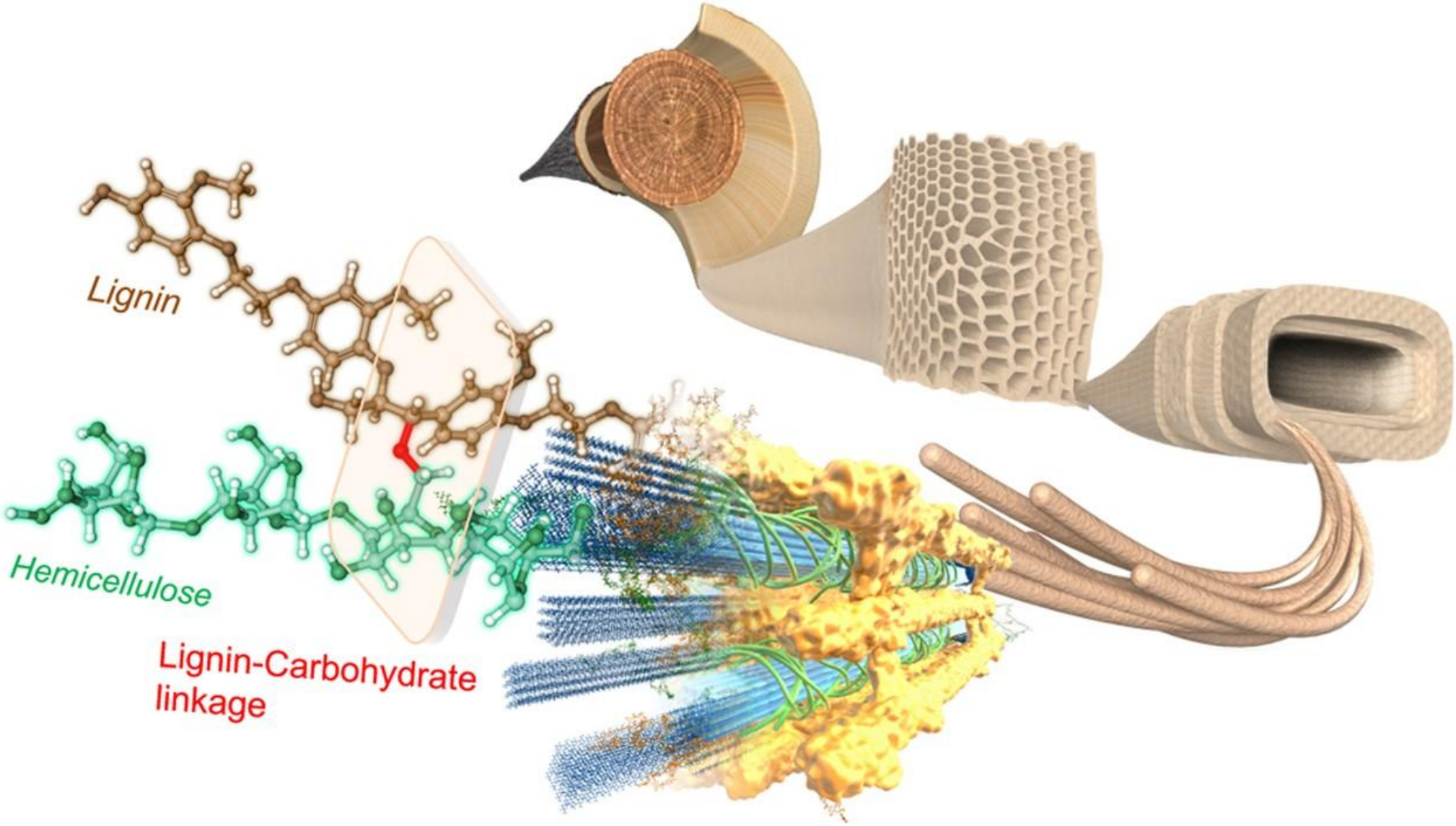
- Flax
- Cotton
- Jute
- Kenaf
- Bamboo
- Ramie
- Sisal

**Woody plants**

- Wood fibres
- Remain alive during dormant season; reinforced by secondary xylem

- Trees
- Shrubs

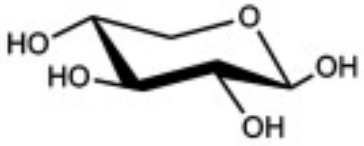
# Plant structure



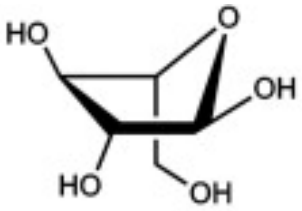

**To get to nanocellulose, this natural composite has to be taken apart.**

# Hemicellulose

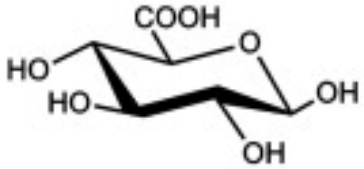
Monomer sugars in hemicellulose



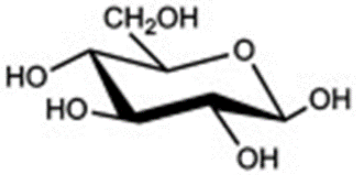
Xylose



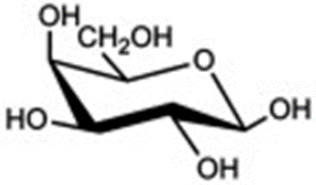
Arabinose



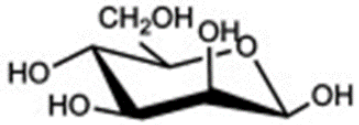
Glucuronic acid



Glucose

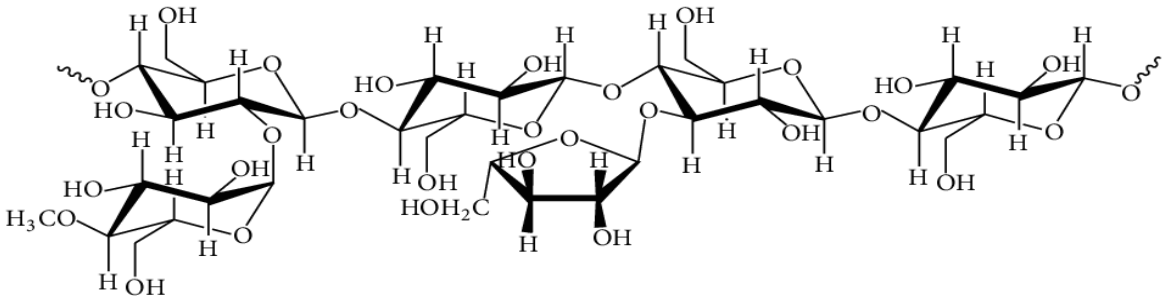


Galactose



Mannose

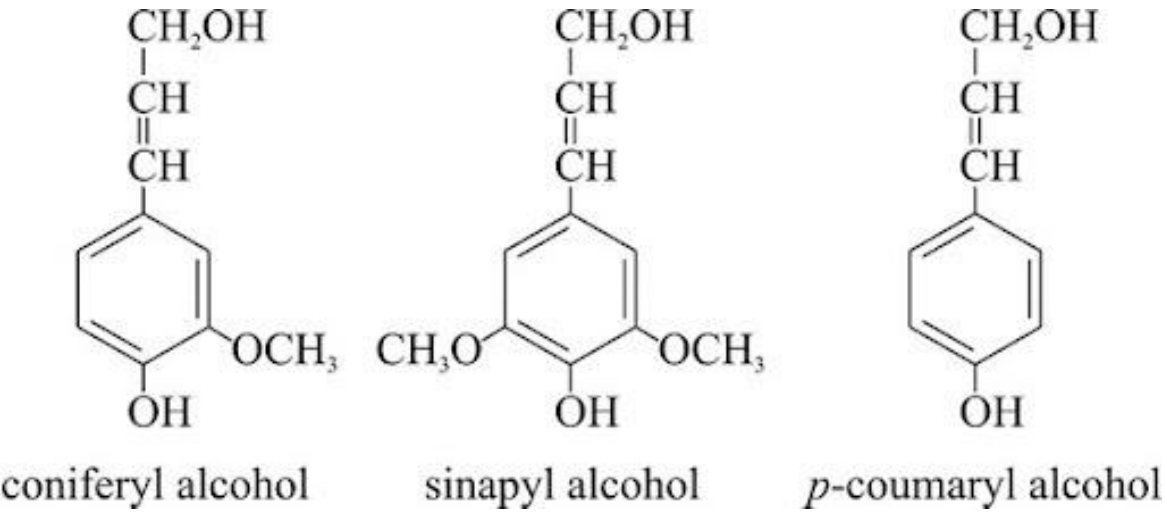
Xylan



**Hemicelluloses are heteropolysaccharides with low DP (<200) and they form amorphous structures in the cell wall**

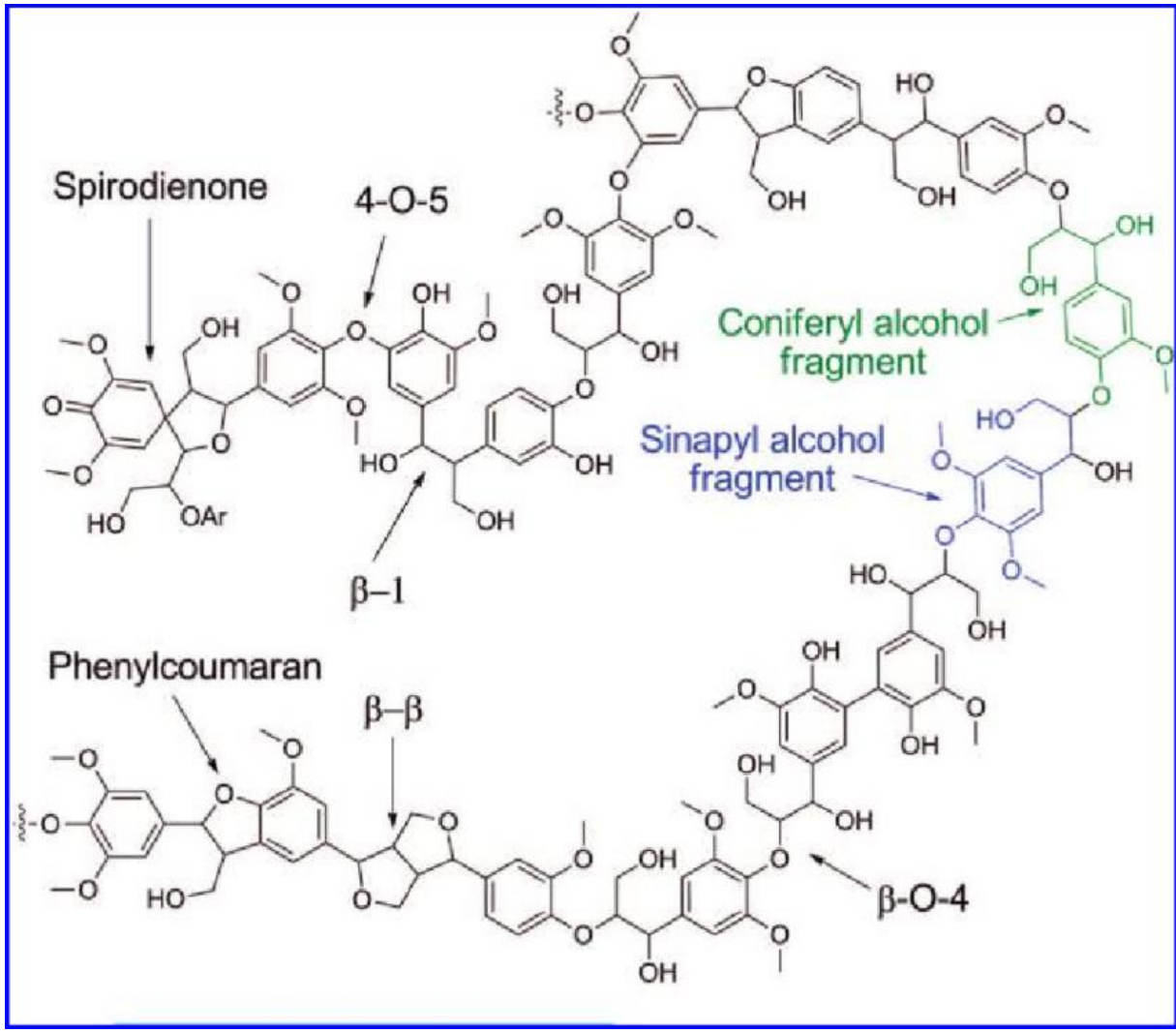
# Lignin

Monolignols




- Non-linear polyphenol
- Structurally extremely diverse
- Glues fibres together as the main component in the middle lamella
- Hydrophobic: controls the amount of water inside the cell wall
- Responsible for the brown colour of the wood

**Hardwood lignin structure**




Applied Microbiology and Biotechnology. 2020, 104, 5201–5212

# Step 1: pulping — isolating the fibers

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**Wood chips or plant fibre**

Raw lignocellulosic material, straight from harvest

→

**Washing, drying & grinding**

Removes surface debris and reduces material to a workable size

→

**Delignification**

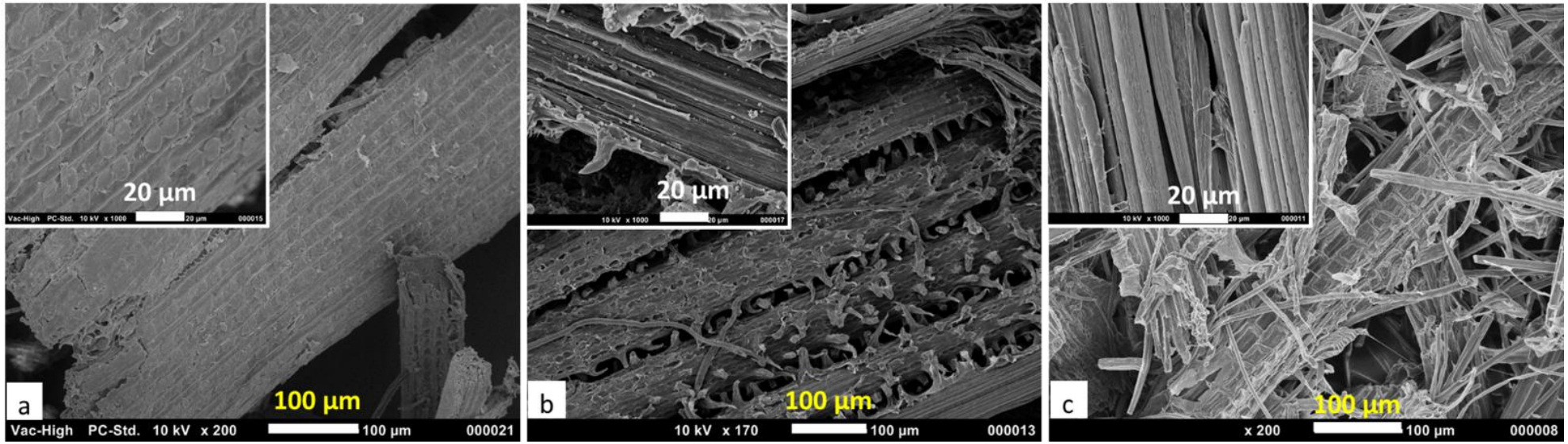
Chemically removes lignin, the rigid “glue” between fibres

→

**Bleaching**

Removes residual lignin and hemicellulose, leaving purified fibre

**Pulping removes lignin and hemicellulose, leaving purified cellulose fibre ready for nanoscale disintegration.**



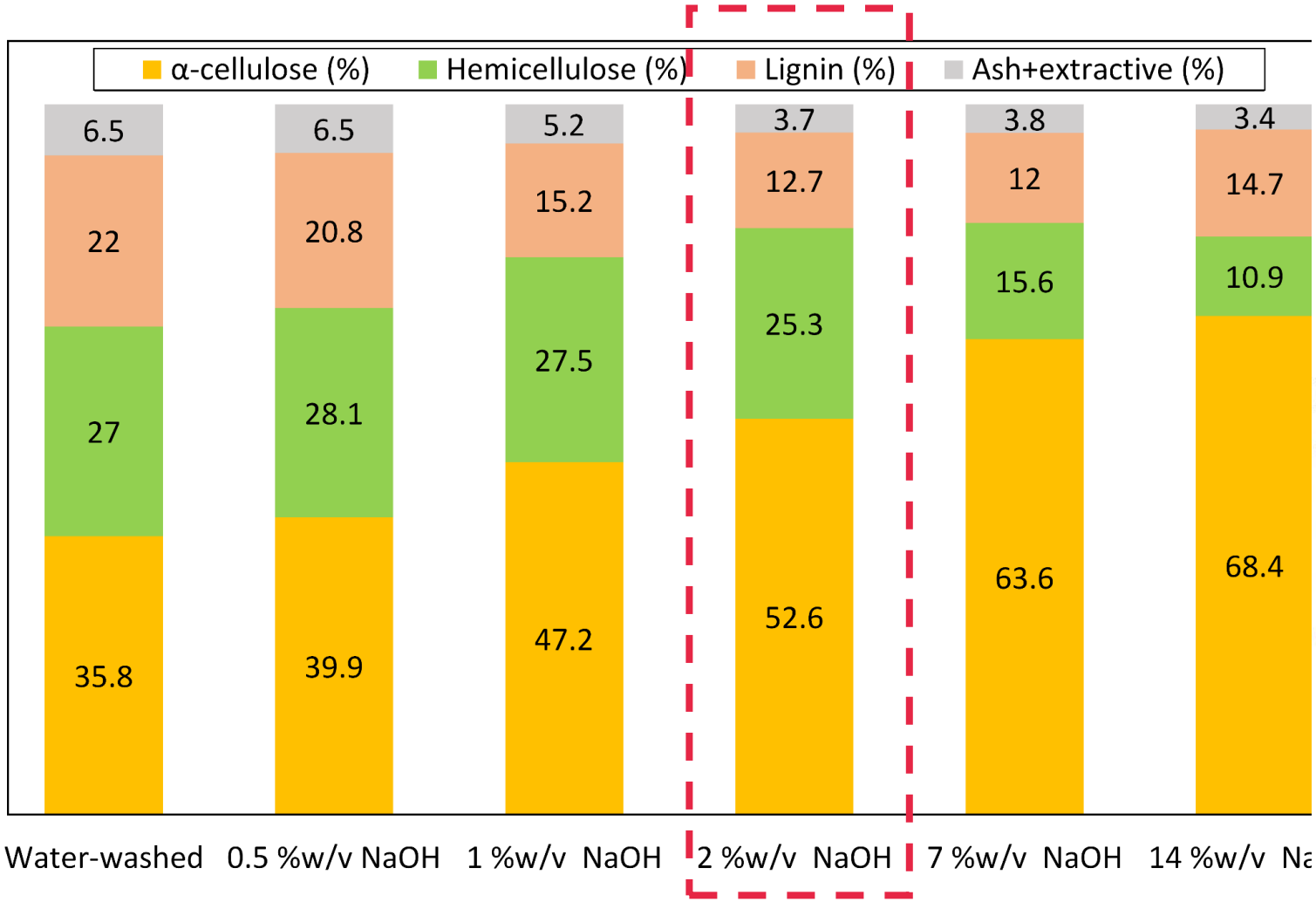

Water washed grass

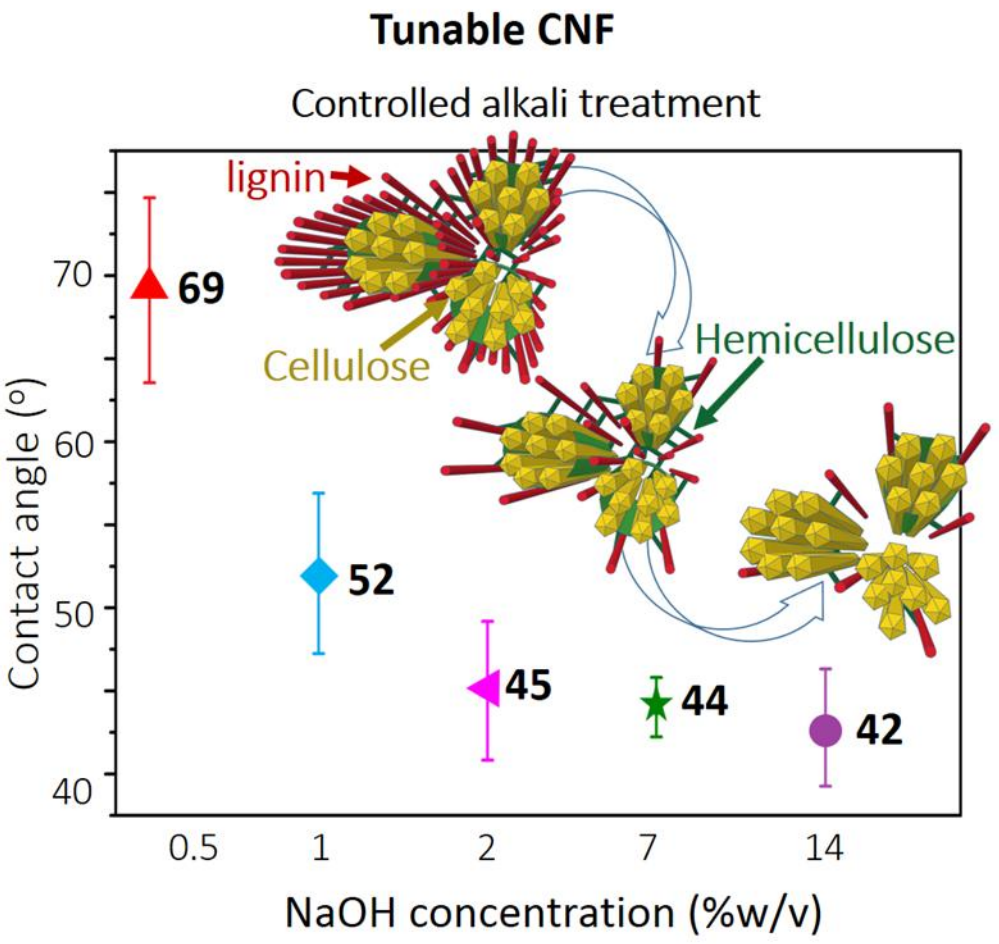
Delignified fibre

Bleached fibre

RSC Advances, 2015, 5 (41), 32124-32132

# Effect of pulping conditions on fibres composition




ACS omega 2019 3 (11), 15933-15942

# Making CNF: from pulp to nanofibre

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**Purified pulp**

From the pulping stage — fibres ready for disintegration

→

**Pretreatment**

Chemical, mechanical or enzymatic — loosens the fibre structure

→

**Mechanical disintegration**

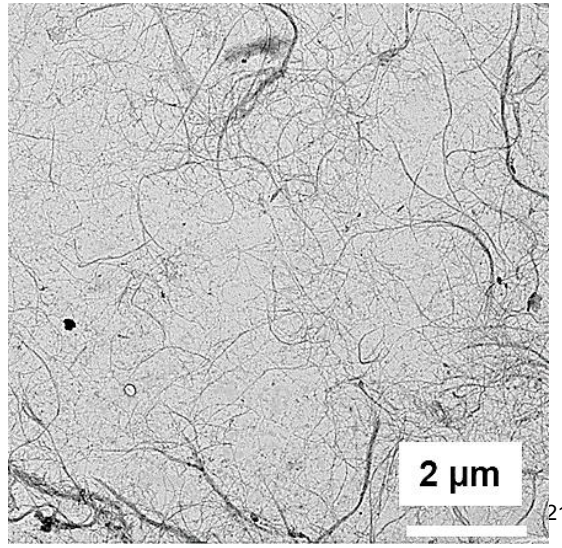
Homogeniser, milling, ultrasonication, Silverson or extrusion

→

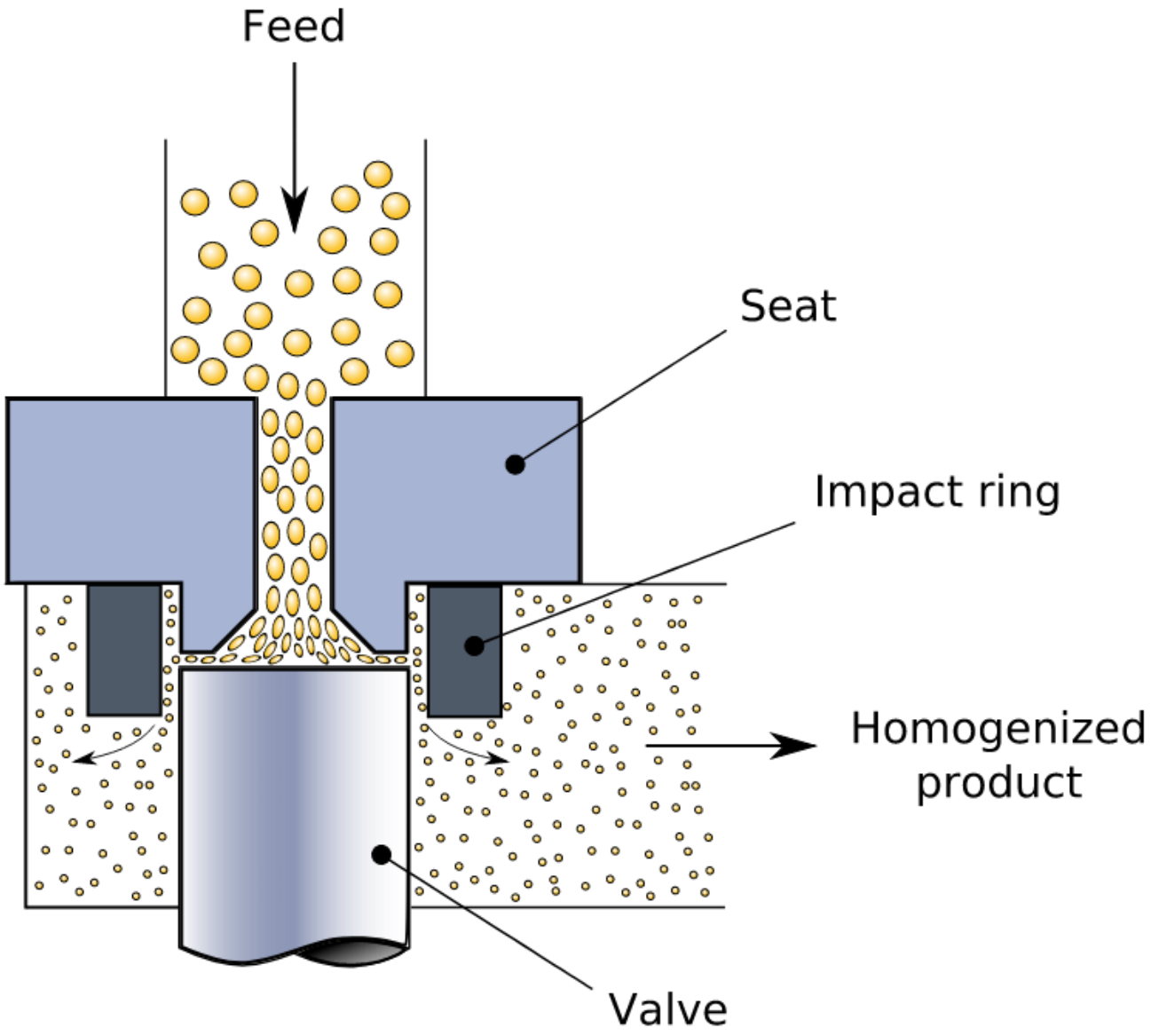
**CNF**

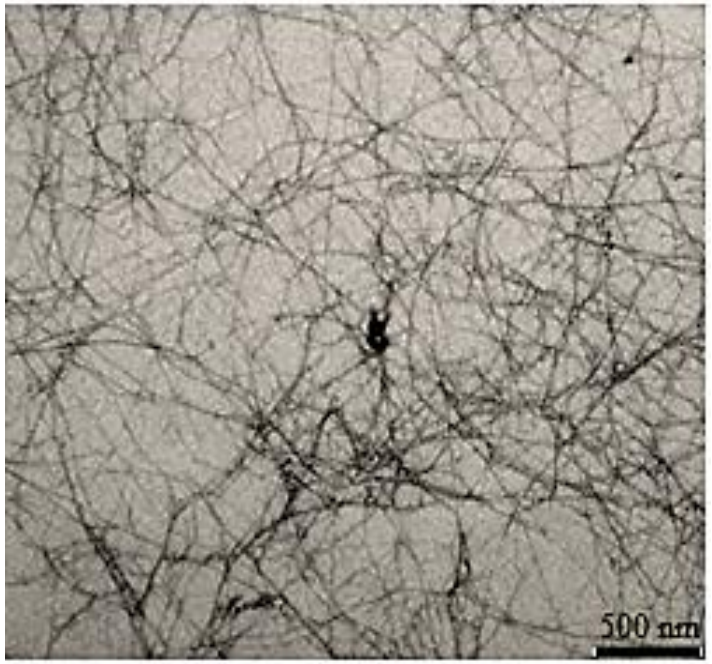
Flexible, high-aspect-ratio cellulose nanofibres

**Several mechanical routes exist — they trade off fibre diameter, energy cost and scalability.**



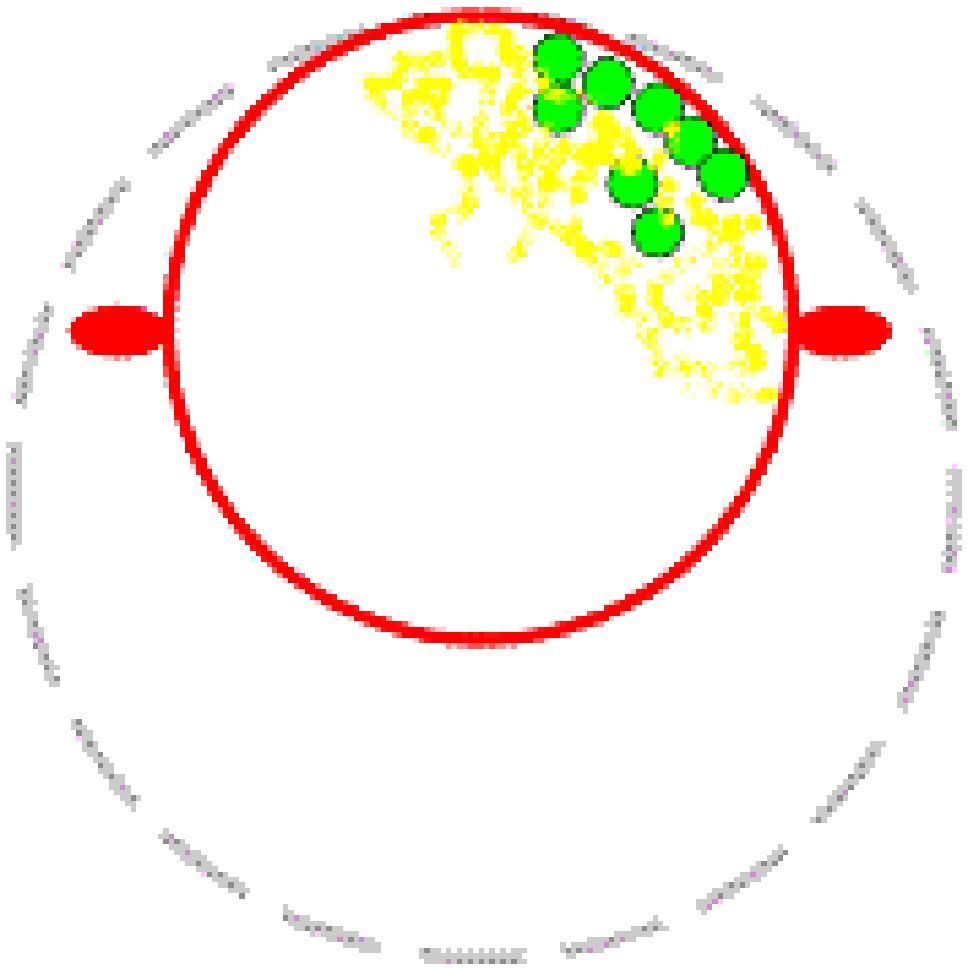
# High pressure homogeniser

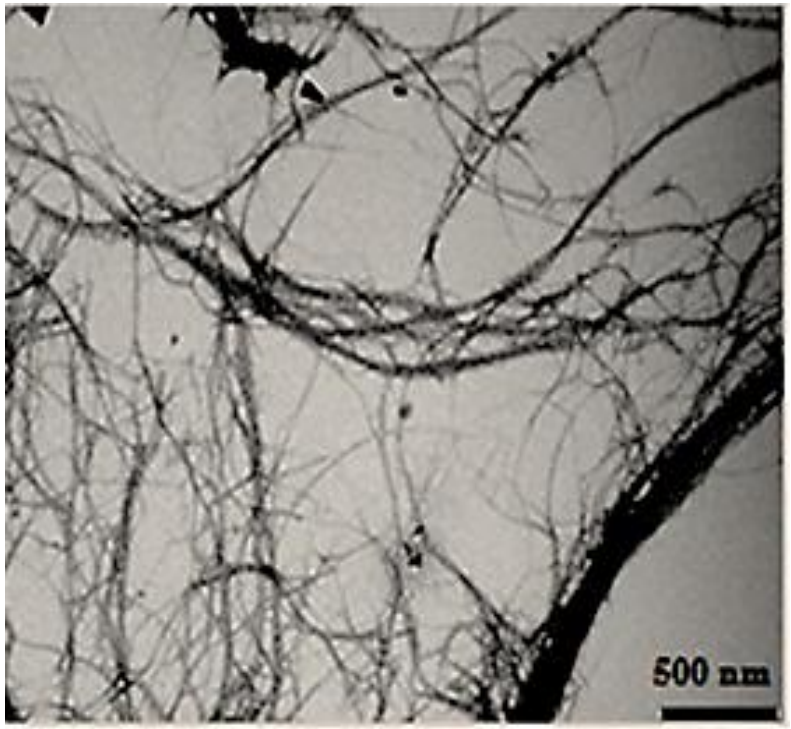


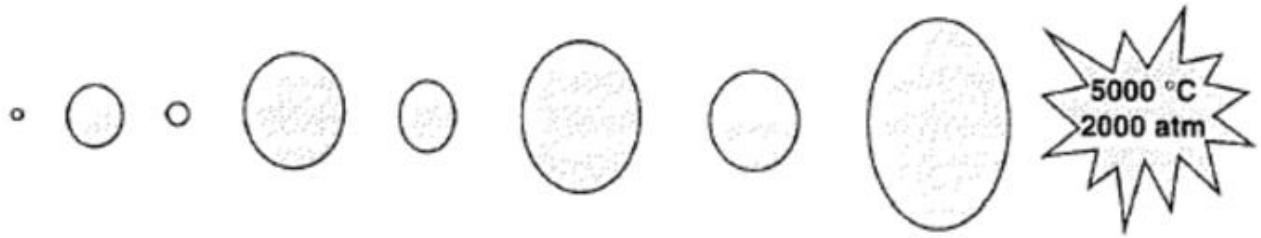

https://www.youtube.com/watch?v=kbyBgLzSSs4

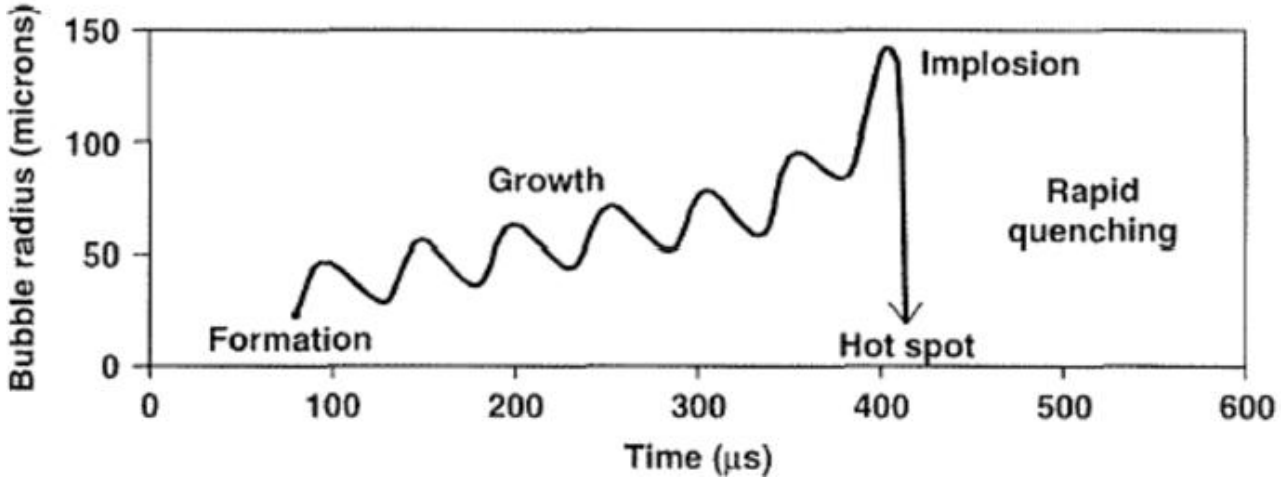
# High energy milling

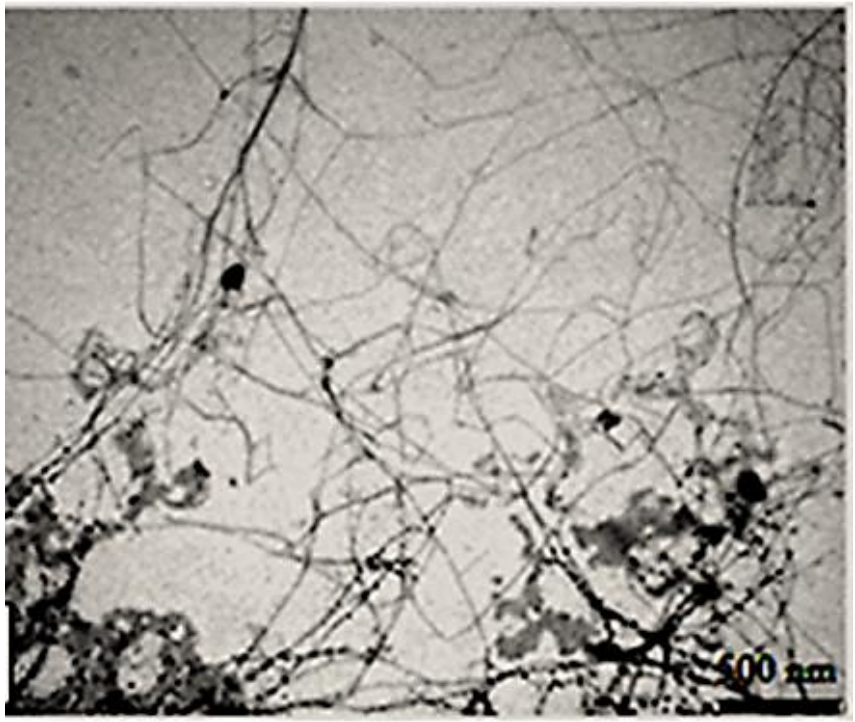


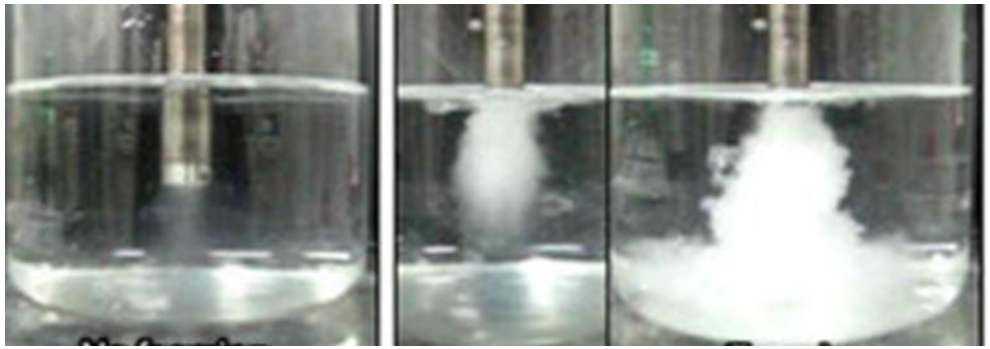


# Ultrasonication



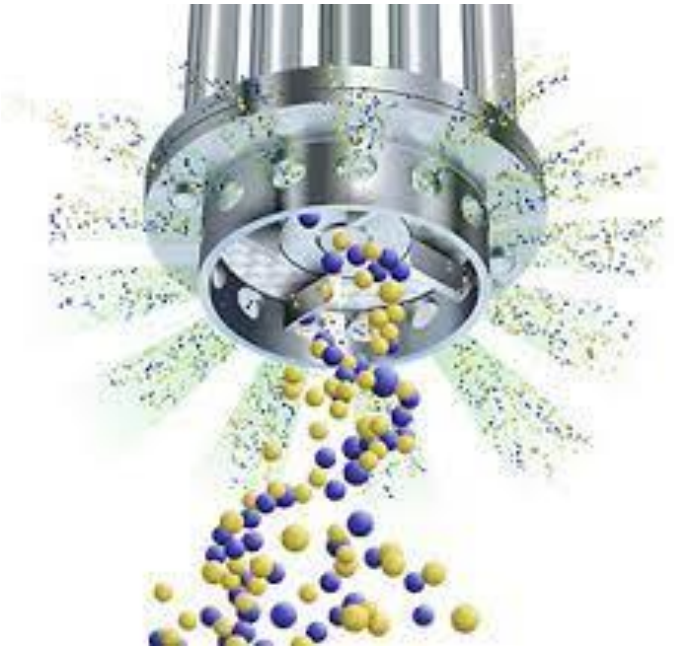
# Silverson

**Stage 1:** The rotor blades inside the special mixing workhead spin really fast and create a strong pull. This pulls liquids and solids from the bottom of the container into the centre of the mixing workhead.

**Stage 2:** Because of the spinning, the materials move to the edges of the mixing workhead. There, they get milled and crushed between the rotor blades and the wall of the mixing workhead.

**Stage 3:** Then, the materials are pushed through tiny holes in the mixing workhead, creating a strong cutting force.

**Stage 4:** The materials that come out of the mixing workhead move outward quickly and hit the sides of the container. At the same time, more material gets pulled into the mixing workhead, so the process keeps going. This movement pattern helps avoid creating bubbles in the liquid.

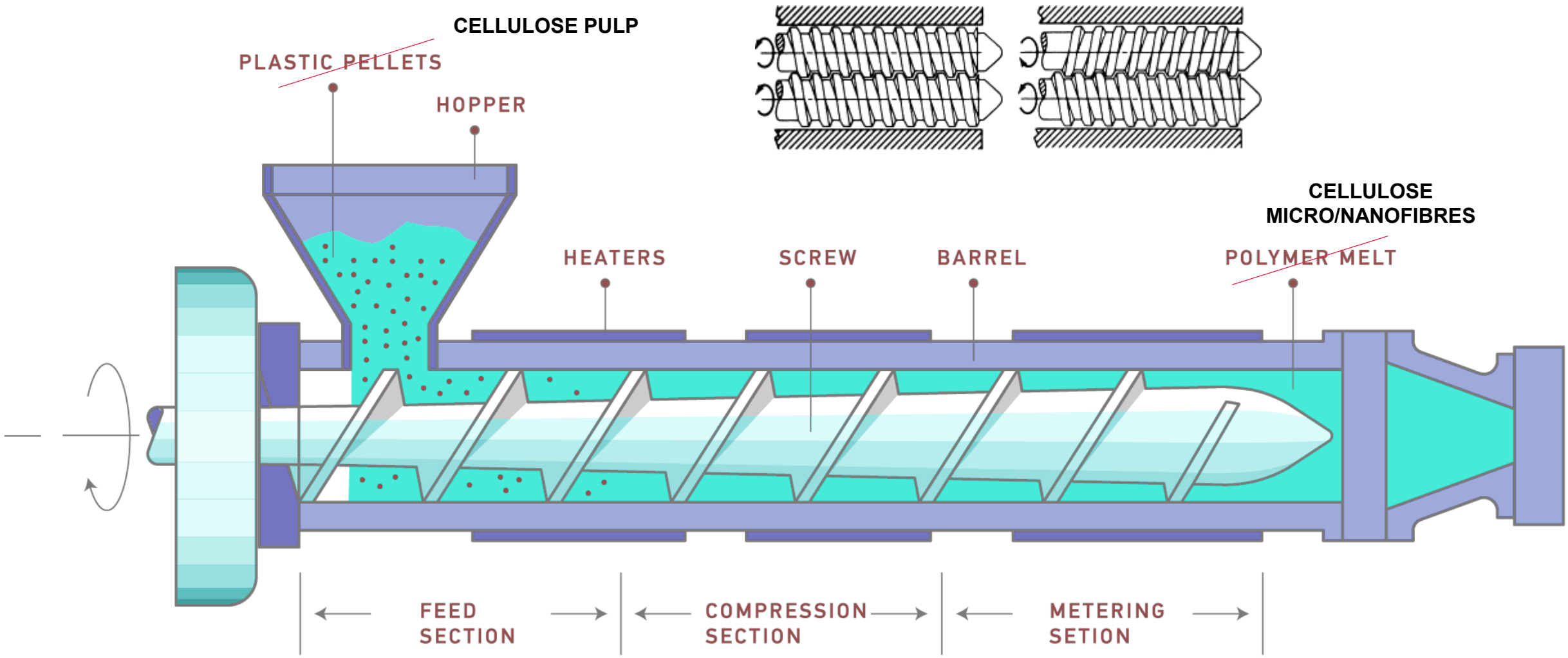






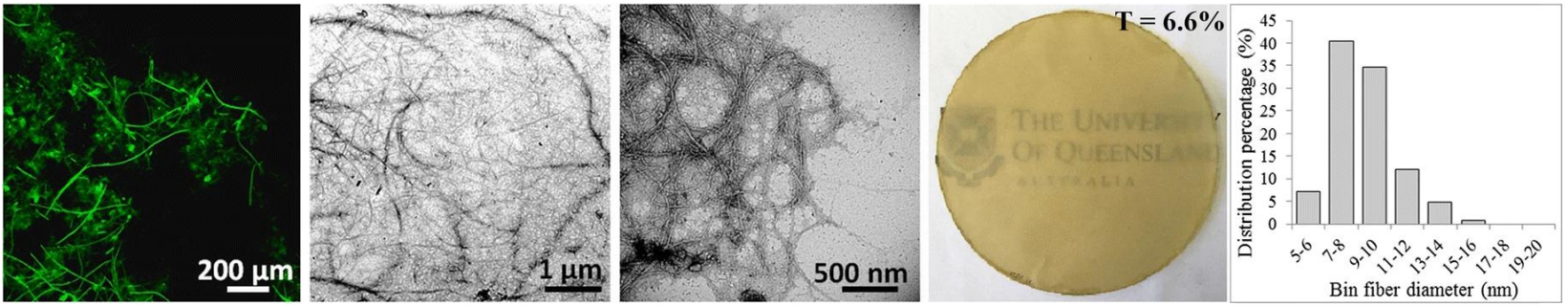

https://www.youtube.com/watch?v=cep81yJlgDk

# Extrusion

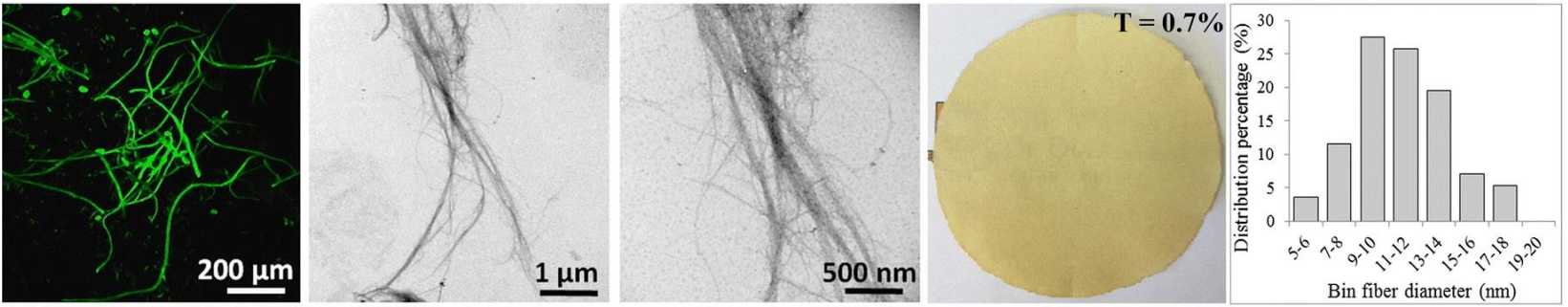



https://www.youtube.com/watch?v=uj-CaGmgsgA

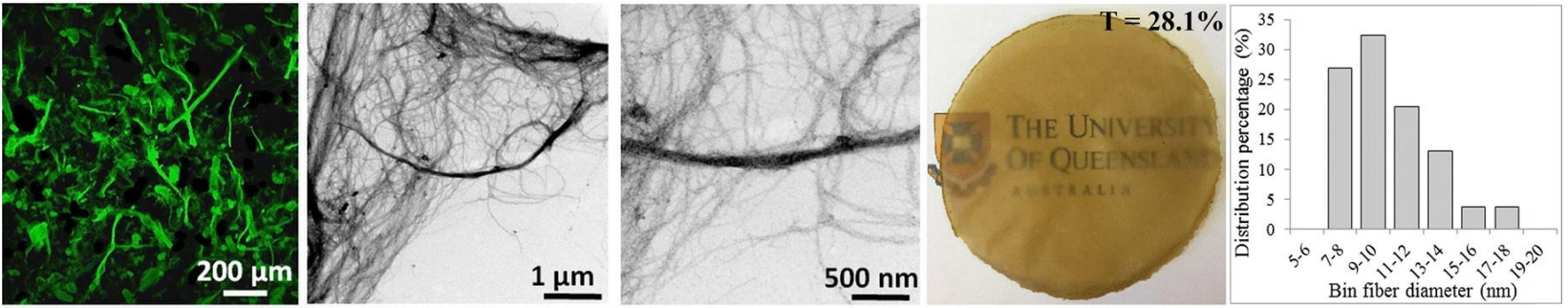
# Nanofibres produced using different methods



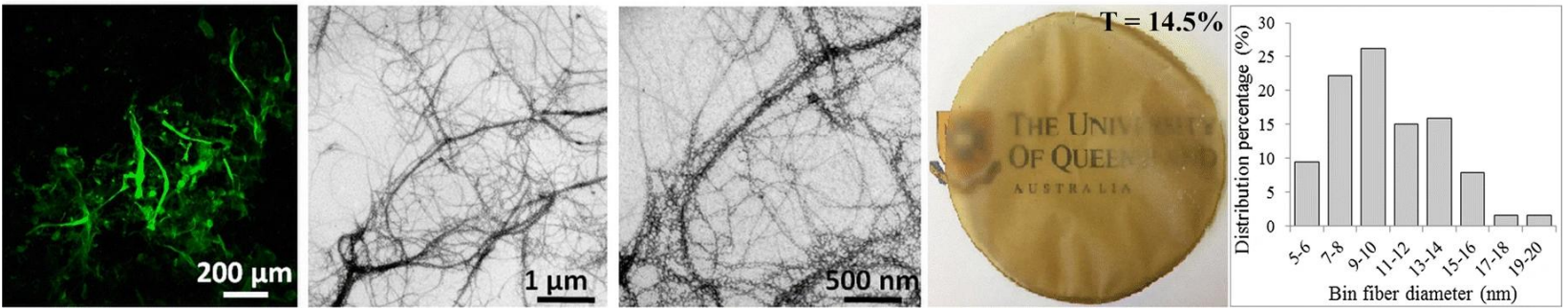

High pressure homogeniser (400,700,3x1100 )
Diameter 8 ± 2 nm




Extrusion ( 3x)
Diameter 11 ± 3 nm




Extrusion (3x) and milling (20 min)
Diameter 10 ± 3 nm




Milling (20 min)
Diameter 10 ± 3 nm

Cellulose, 26, 6555–6569 (2019)

# Making CNC: acid hydrolysis

**Purified pulp**

From the pulping stage — fibres ready for hydrolysis

**Acid hydrolysis**

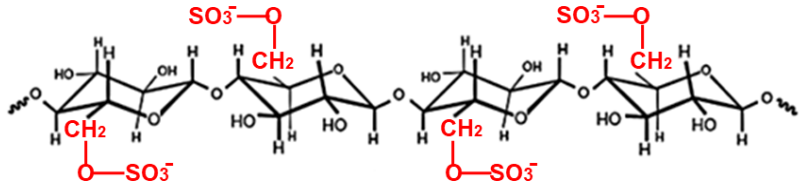
Sulphuric, hydrochloric, phosphoric, or nitric acid + sodium nitrate

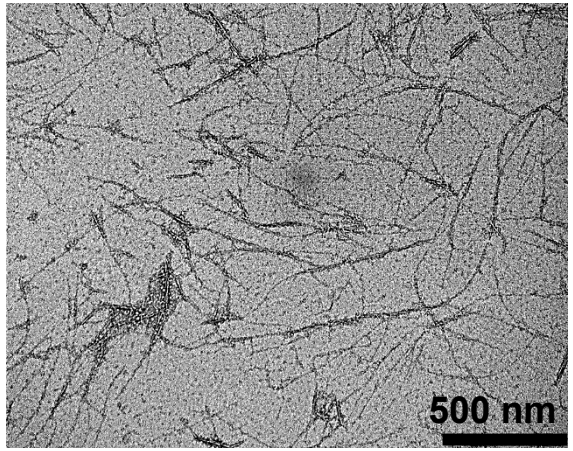
**Centrifuge & dialysis**

Removes excess acid and reaction by-products

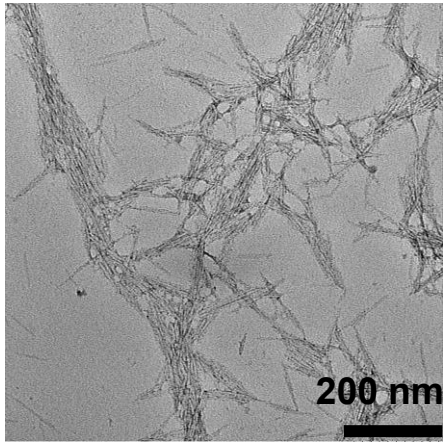
**Ultrasonication → CNC**

Disperses the crystals into rod-like cellulose nanocrystals

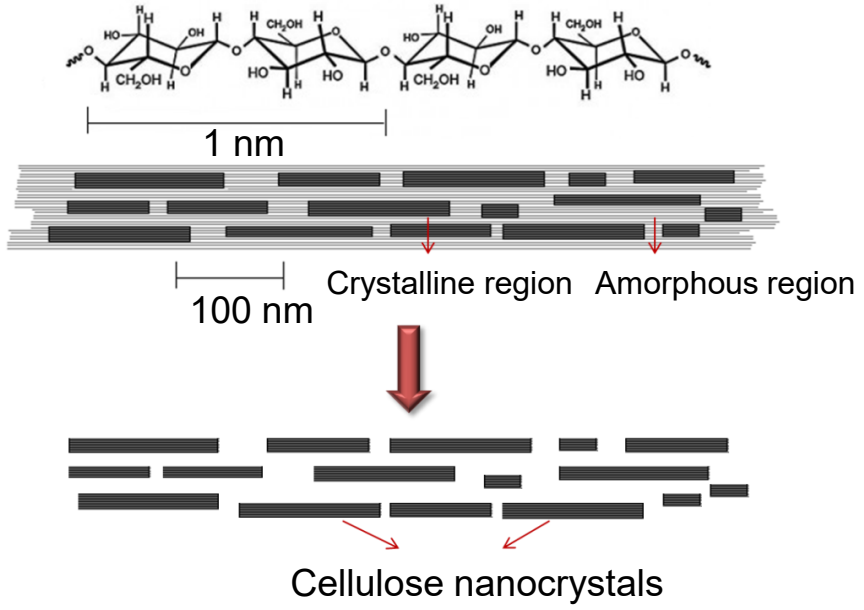



Mild acid treatment of grass




Acid treatment of wood




Cellulose, 2017 24 (9), 3753-3766

# Different types of nanocellulose – a comparison

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## Cellulose nanofibers

- Gels at low concentrations (0.5-1.5 w%)
- Very high aspect ratio (l/d can be ~1000)
- Flexible, spaghetti-like shape
- Behaviour is not always well defined
- Intensive research from 2007 onwards
- Promising in especially composites applications (reinforcing phase)

## Cellulose nanocrystals

- Fluid dispersion at low concentrations
- Not very high aspect ratio
- Rod-like: always straight shape
- Intensive research from 1992 onwards (although a surge after 2005 occurred)
- Promising as filler material, viscosity controller etc.
- Many high-end applications have been proposed

# Nanocellulose characterisation

**Morphology**: Transmission electron microscopy (TEM)
Scanning electron microscopy (SEM)
Dynamic Light Scattering (DLS)

**Surface charge**: Zeta potential

**Surface area**: BET

**Degree of crystallinity**: X-ray diffraction (XRD)

**Thermal stability**: Thermogravimetric analysis (TGA)
Differential scanning calorimetry (DSC)

**Mechanical properties**: Bulk nanofibers: Rheology, Instron
Individual nanofiber: AFM

**Surface functionality**: Fourier transform infrared (FTIR)
X-ray photoelectron spectroscopy (XPS)

**Lignocellulose composition**

# Why nanocellulose is useful for sustainable materials

**High aspect ratio**

reinforcement and network formation

**Hydroxyl-rich surface**

functionalisation and hydrogen bonding

**Film-forming ability**

dense oxygen-barrier layers when dry

**Rheology modification**

stabilisation of coatings, inks and emulsions

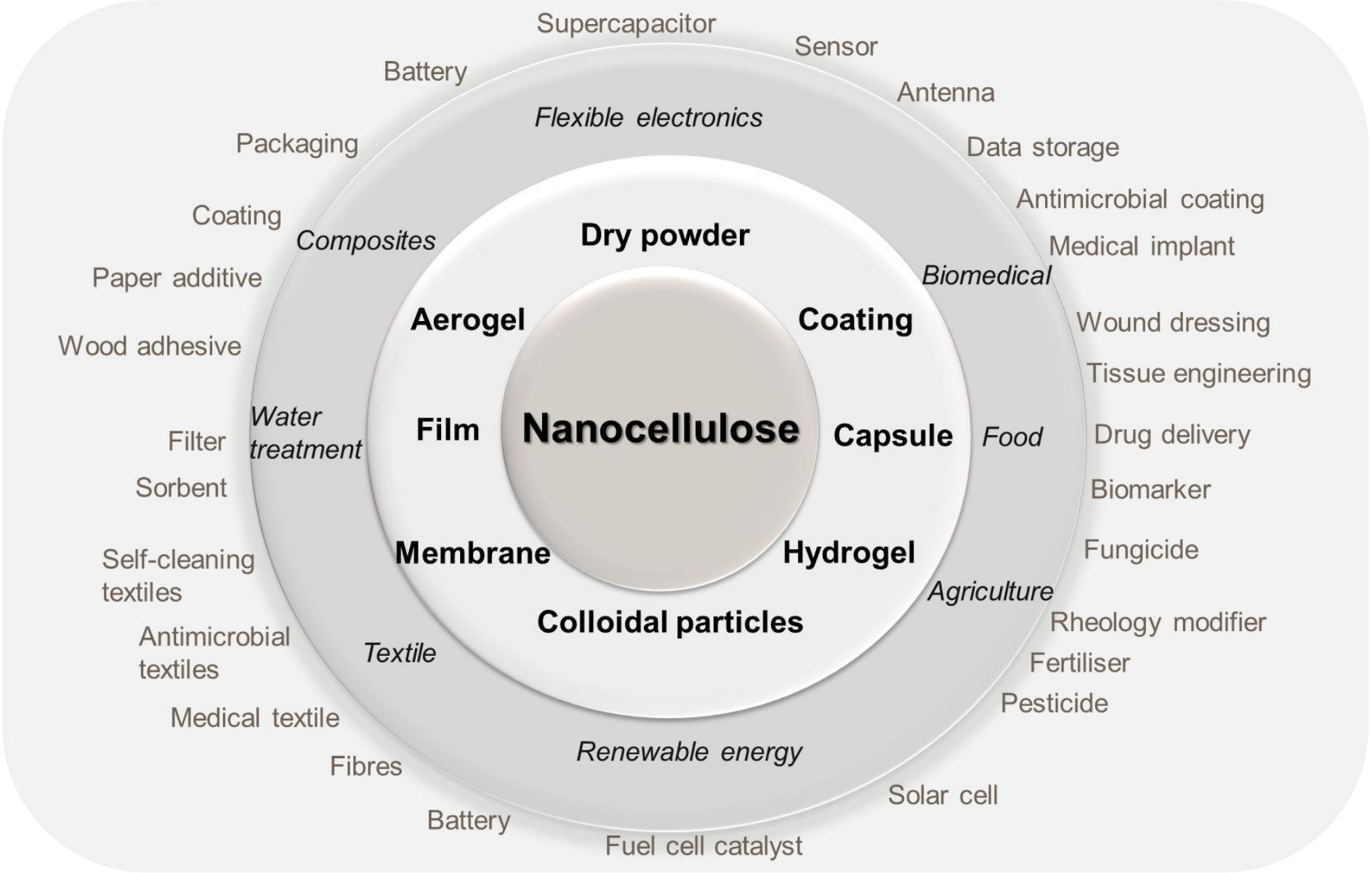
**Bio-based origin**

renewable carbon input — not automatically sustainable

**Surface tunability**

antimicrobial, hydrophobic or low-fouling functions

# Why industry cares about nanocellulose



# Challenges

High surface area and high surface energy lead to agglomeration

Not compatible with hydrophobic polymers and non-polar solvent

Moisture sensitivity

Drying and shipping cost

**Most of today's research effort goes into solving exactly these four problems.**

# Surface modification: the engineering answer

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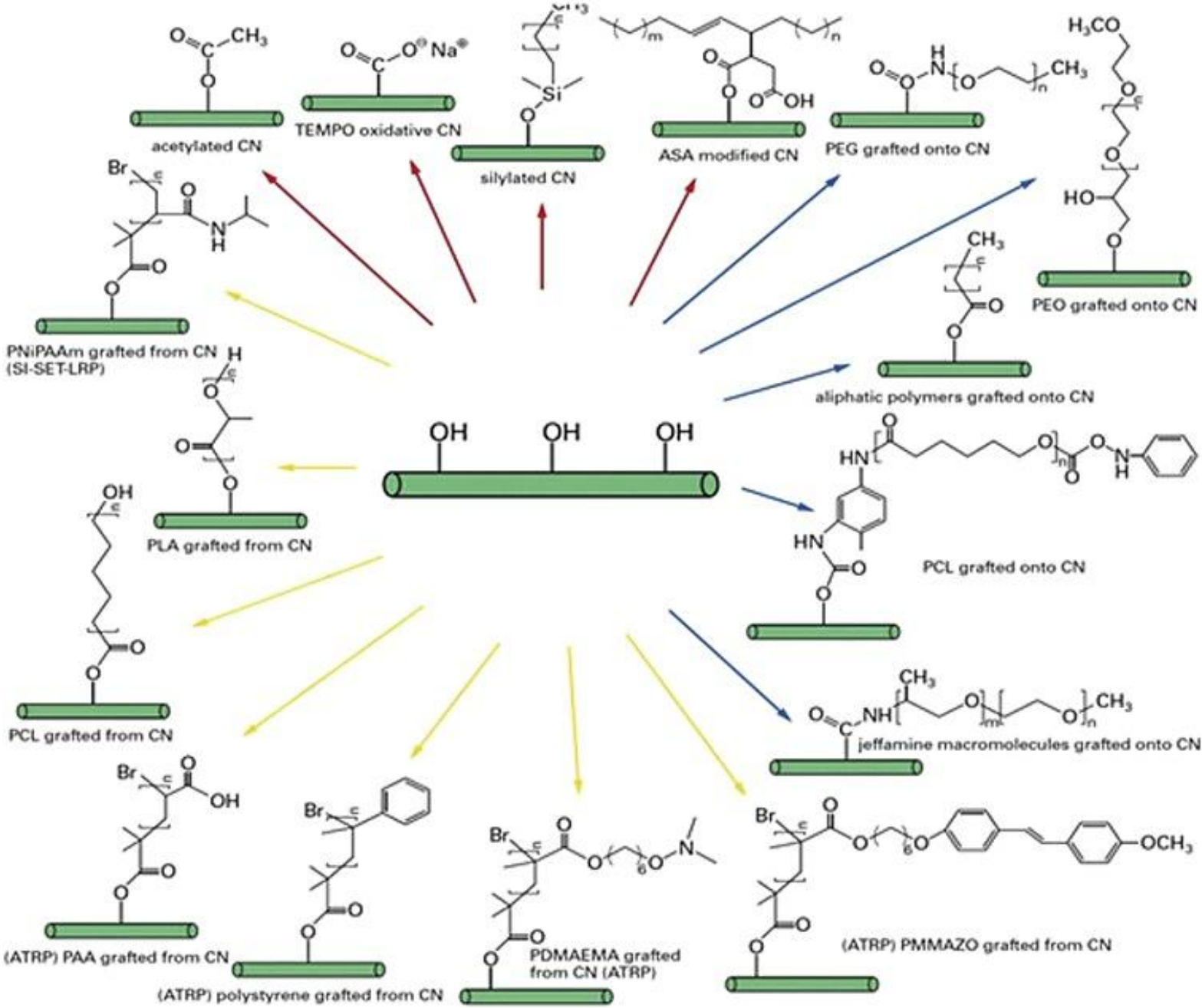
## Adsorption methods

## Chemical methods

- Silylation
- Acetylation/Esterification
- Carbonylation
- Functionalised CNF reactions

## Modification by polymer grafting

**Each approach trades off cost, scalability, and how much of the native fibre properties survive.**




Biotechnology Reports, 29, 2021, e00582

# Nanocellulose-based commercial products

| Company / product | Type | Commercial application |
|---|---|---|
| **CelluForce NCC® / CelluShield™** | CNC | Barrier coatings and performance additives for packaging and other formulated products. |
| **Borregaard Exilva®** | MFC | Rheology modifier and stabiliser in paints, coatings, adhesives, detergents, and personal care. |
| **FiberLean MFC** | MFC | Paper, packaging, tissue, and speciality fibre applications. |
| **UPM GrowDex® / FibDex® / FibGel™** | NFC | 3D cell culture hydrogel, wound care, and medical-device hydrogel applications. |
| **Nippon Paper Cellenpia®** | CNF | Food, cosmetics, rubber, and industrial material applications. |

Most nanocellulose products are not sold directly to consumers; they are enabling materials used to improve function, sustainability, and processability.

# Existing commercial applications of nanocellulose




Adult diaper
(CNF, Japan)



Viscosity control in
oil drilling
(CNC, Canada)



Ink dispersing in
ball point pen
(CNF, Japan)




CelluShield™-
barrier coating

**Multi-tonne pilot and production facilities are already running across North America, Japan and Europe.**

# Case study: from spiky grass to nanocellulose

*From material extraction to community-led biomass valorisation*

# Years of knowledge – the world's longest-running R&D program

## Country and knowledge systems

- Aboriginal and Torres Strait Islander peoples have developed and refined knowledge of Australia's unique flora, fauna, and materials for over 100,000+ years.
- This knowledge is not 'folklore' – it is a sophisticated empirical system developed through rigorous observation, intergenerational transmission, and adaptive management.
- Plant-derived materials, natural antimicrobials (e.g. tea tree oil, eucalyptus), and fibre processing techniques are part of this knowledge base.
- This knowledge has high relevance to biomanufacturing – but engaging with it ethically is non-negotiable.

## Why this matters for biomanufacturing

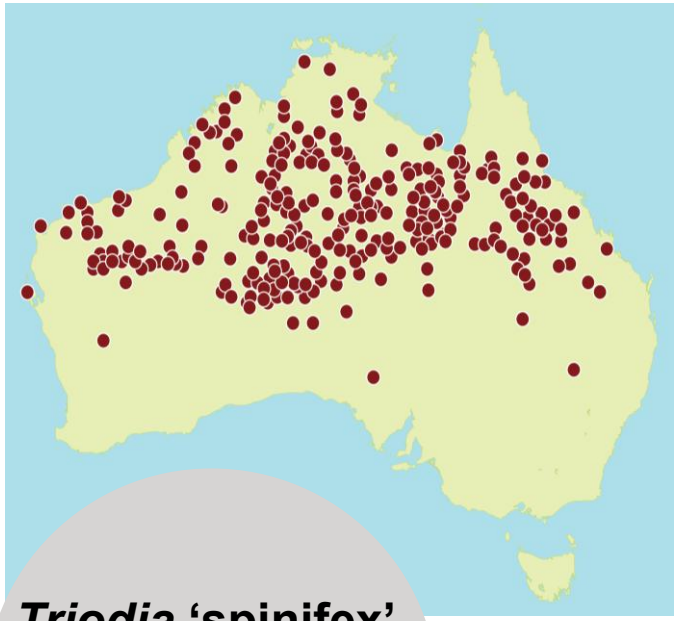
- Traditional knowledge can identify promising feedstocks and applications faster than systematic screening.
- Community partnerships enable access to land and materials under ethical frameworks – not extraction.
- Indigenous communities are increasingly asserting sovereignty over biological resources and associated knowledge.
- A biomanufacturing researcher who does not understand this context will make costly ethical and legal errors.

# Principles of ethical research collaboration

Five principles are drawn from participatory action research frameworks used in Australian Indigenous health and environmental research, and from the Nagoya Protocol on access and benefit-sharing:

| | |
|---|---|
| **Community-led** | Research questions emerge from community priorities, not only from lab priorities. The community defines what is useful, not just what is scientifically interesting. |
| **Two-way knowledge** | Both knowledge systems – Western scientific and Indigenous – contribute, and both are respected. Results are shared back with the community in accessible, plain-language formats. |
| **Free, Prior & Informed Consent** | No material collection, no knowledge documentation, no publication without explicit consent from the appropriate knowledge holders. This is an ongoing relationship, not a one-time agreement. |
| **Benefit sharing** | Economic and non-economic benefits flow back to the community – capacity building, employment, IP co-ownership where relevant, community-identified outcomes. |
| **Material Transfer Agreements** | Formal legal agreements govern what plant or biological material can be taken, for what purpose, and with what restrictions on further use or commercialisation. |

# From spiky grass…



***Triodia* 'spinifex'**

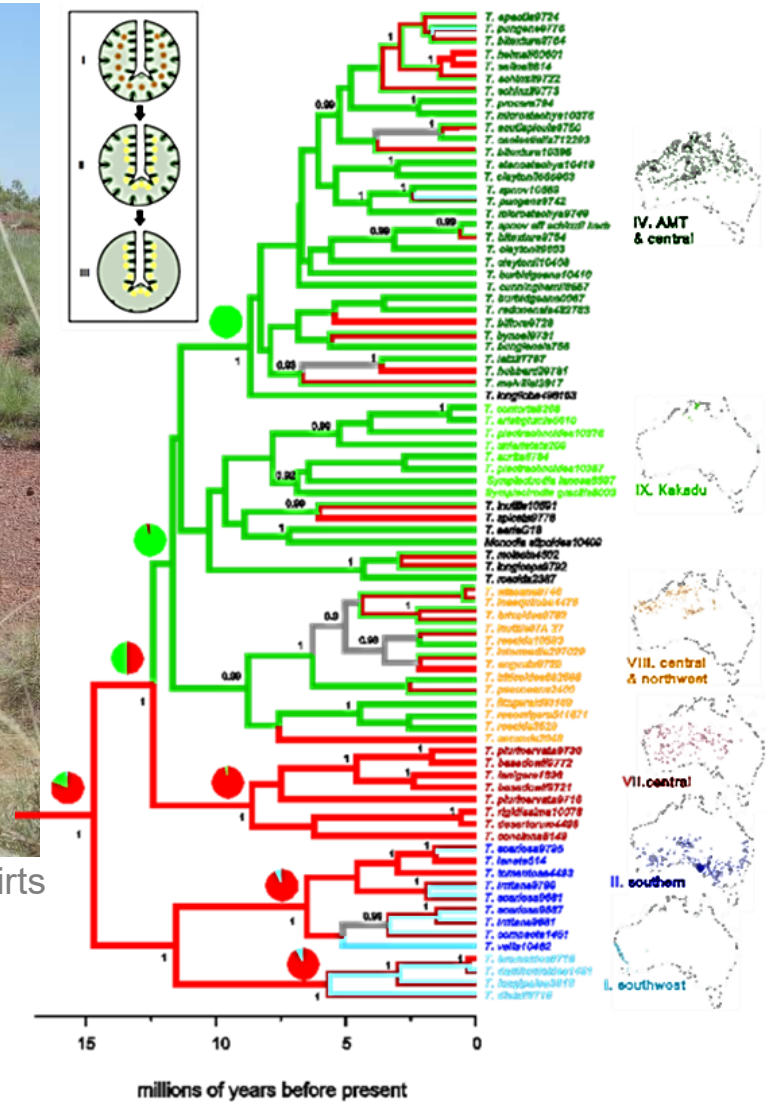
**Covers >2 mln km$^2$**

**~ 69 species**



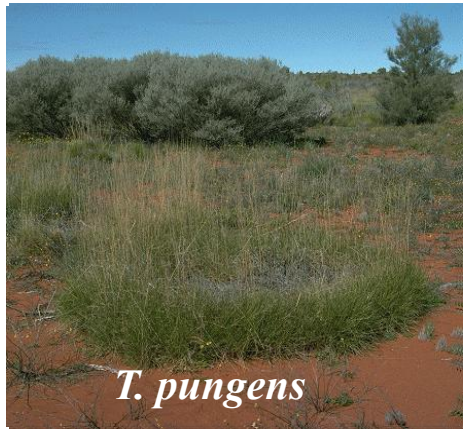
**Extremophile, abundant in remote, arid zones of Australia**

Spinifex harvesting in the outskirts of Camooweal, QLD



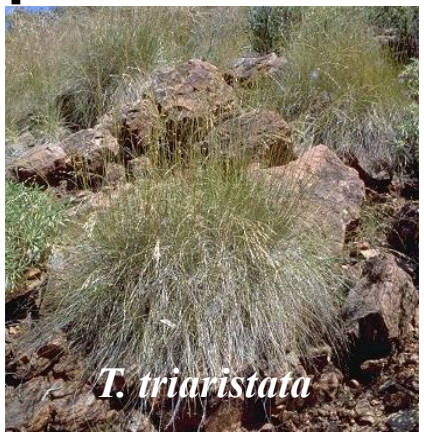
# Some species of spinifex

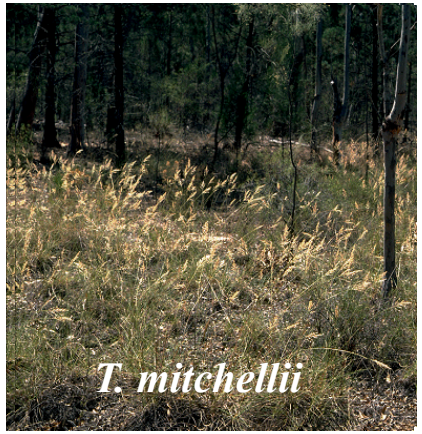
## Soft spinifex





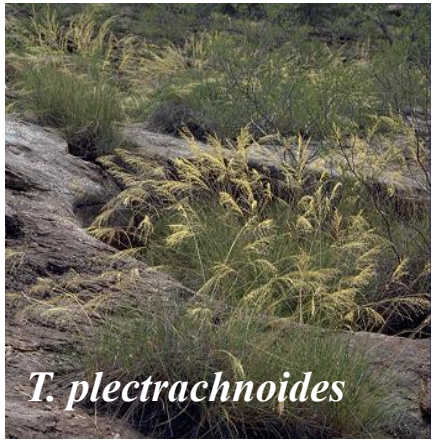


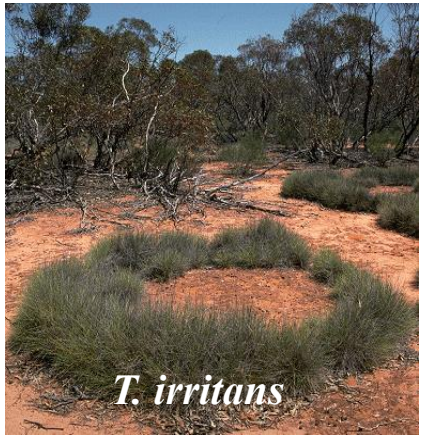


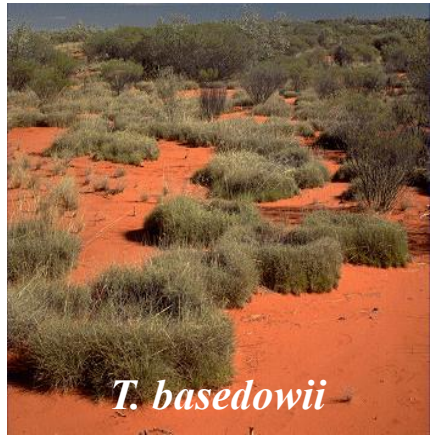




## Hard spinifex



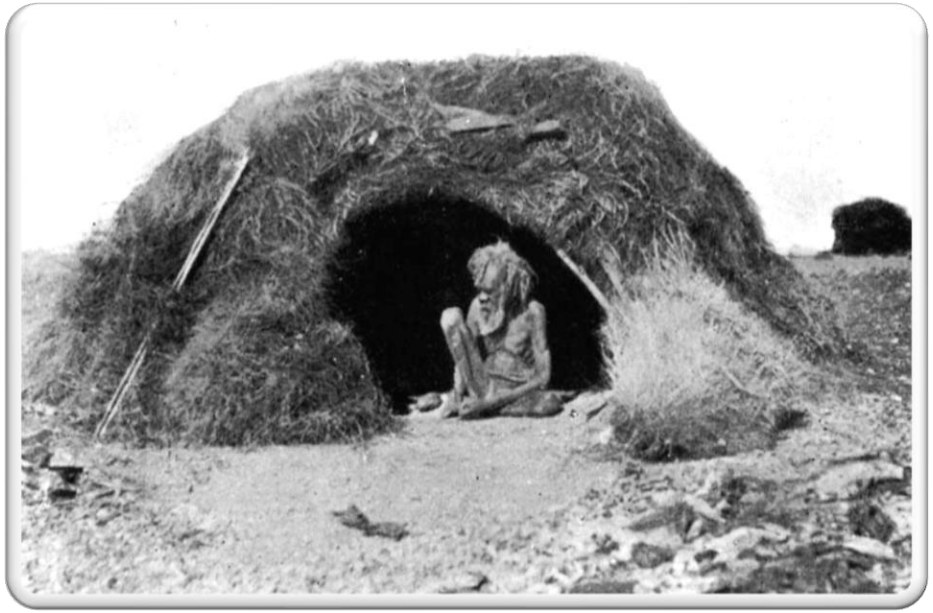

Photos: Aus Grass CD

# Traditional applications of spinifex





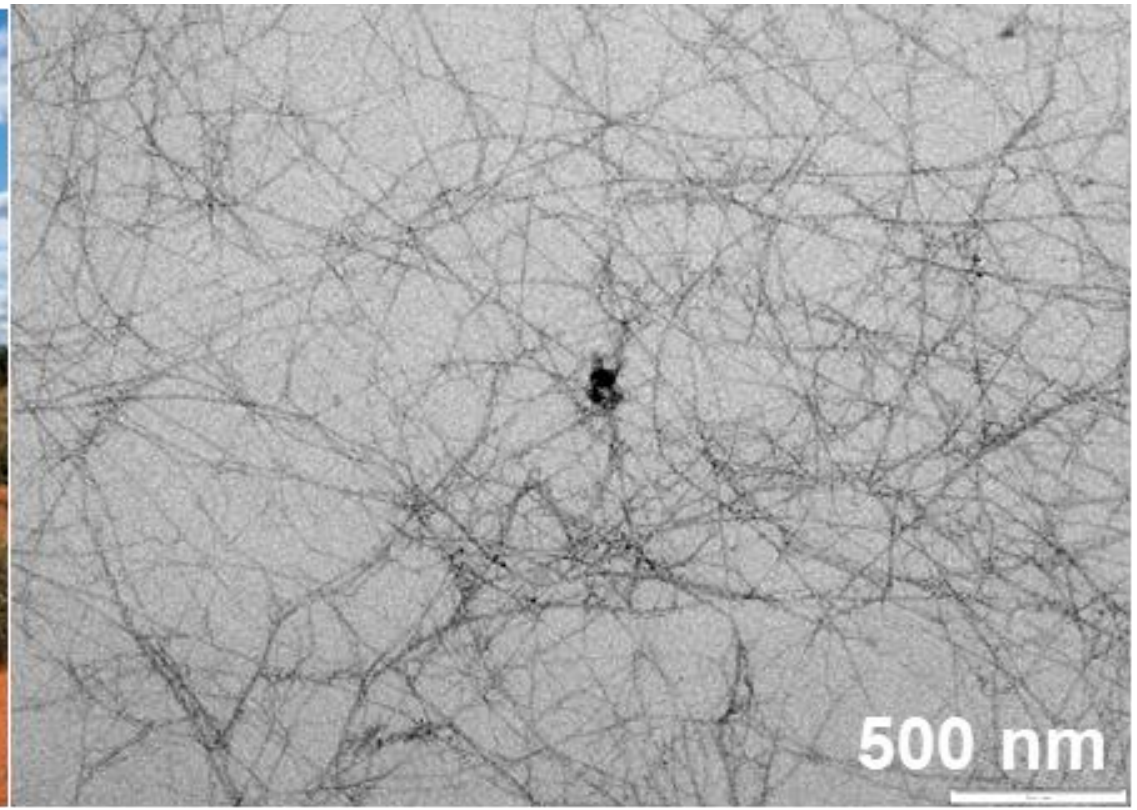




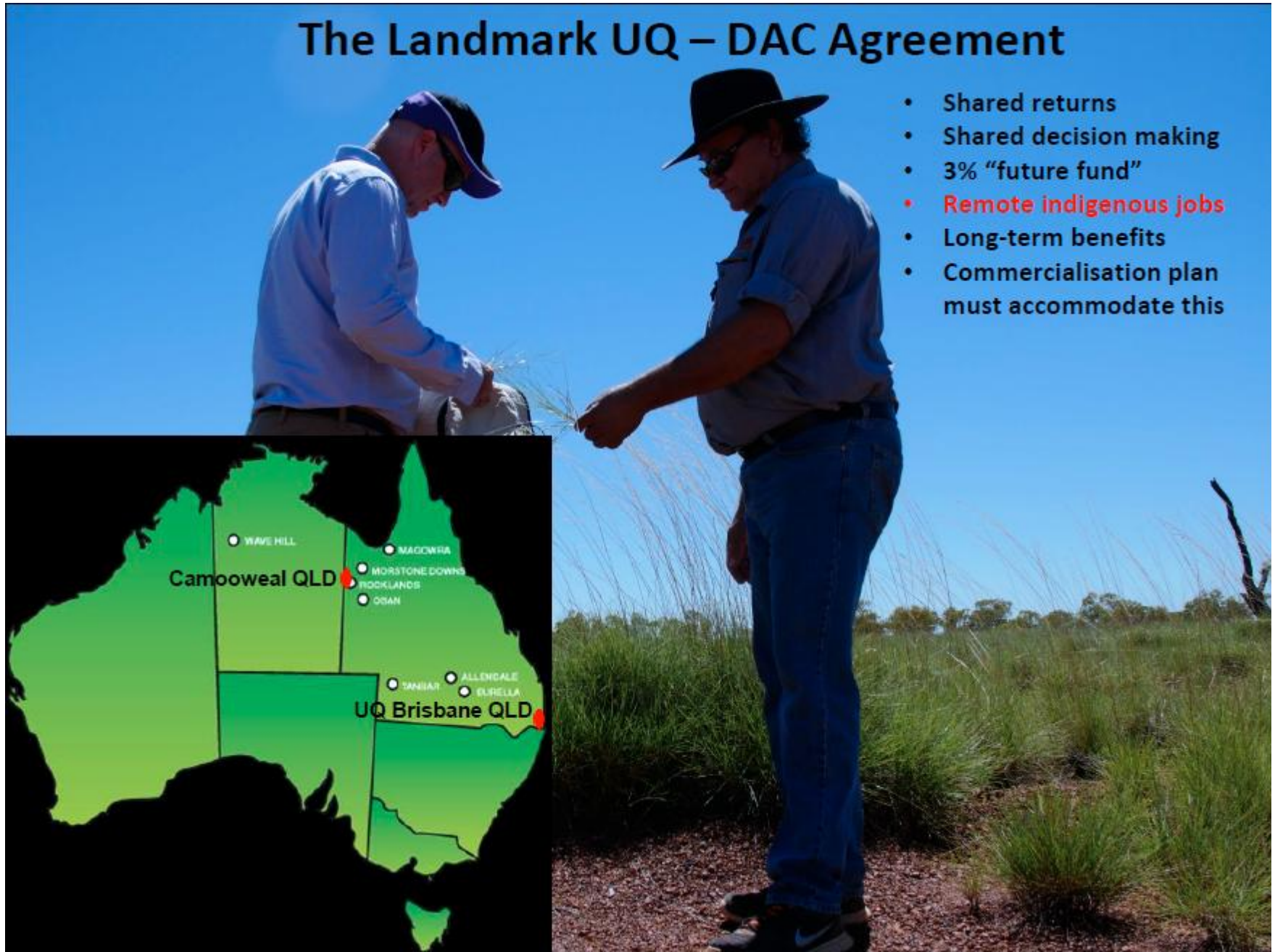


# Spinifex cellulose nanofibers





# Spinifex nanotechnology



# Spinifex nanocellulose technology

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- ✓ Innovate solutions to solve society's problems
- ✓ Sustainable materials
- ✓ Translational progress



Cement and concrete

**Australia's first Nanocellulose pilot plant facility**

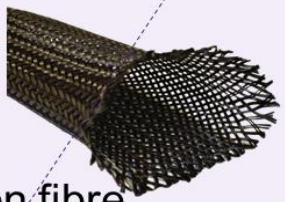





Water filtration



Bitumen



Carbon fibre



Dry compou

**Triodia Wilingi**



ard



AU Patent WO2015074120 A1
AU Patent WO2016090425

# What changed because it was a partnership?

The partnership influenced not only who participated, but how the technology was designed, governed and translated.

### Research design

The research questions, materials focus and intended applications were shaped through partnership rather than defined only by the laboratory.

### Governance

Access to biological resources, knowledge use, consent and decision-making required explicit agreement rather than being treated as informal background issues.

### Capability building

The work was linked to skills, participation and enterprise opportunities, not only publication outputs.

### Value creation

The pathway was considered in terms of who benefits, how value is shared and whether local manufacturing or community outcomes are possible.

### Long-term accountability

Success was judged not only by the invention itself, but by whether relationships, responsibilities and benefits could be sustained over time.

**Partnership changed the translation pathway, not just the source of the biomass.**

# Shift the model: from extractive research to co-designed value creation

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AUSTRALIA

**Extractive model**

Researcher defines problem

Community provides material or access

Benefits captured mainly by institution

Limited long-term accountability

→

**Co-designed model**

Community defines priorities with researchers

Governance, IP and benefit are agreed early

Capability and local value are considered

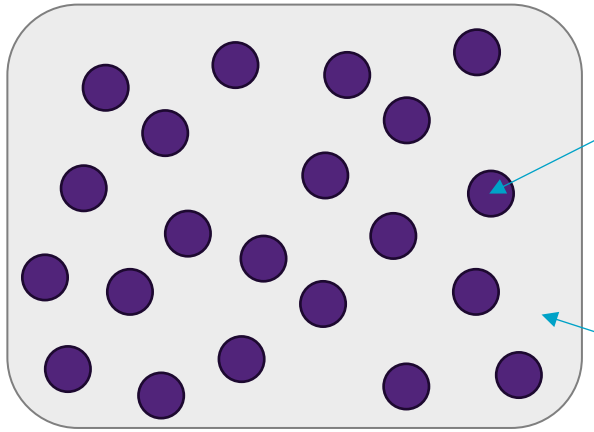
Long-term accountability is explicit

# Case study: from spiky grass to nanocellulose

*Reinforcing natural rubber latex for medical gloves*

# Nanocomposites from nanocellulose

Schematic cross section
of a nanocomposite



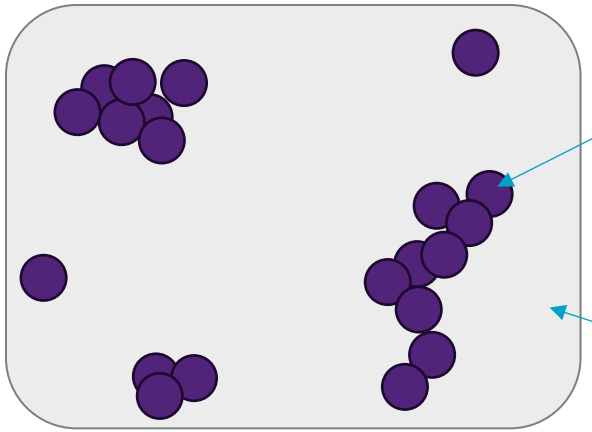
Reinforcing phase: Nanofibril or nanocrystal cross section (cellulose)

Continuous polymer matrix (preferably from a biodegradable polymer)

WHY NANOCOMPOSITES?
Nanocellulose has a very large surface area, that is, many contact points with the continuous polymer matrix, plus it is very strong.

# Nanocomposites from nanocellulose

Schematic cross section
of a nanocomposite



Reinforcing phase: Nanofibril or nanocrystal cross section (cellulose)

Continuous polymer matrix
(preferably from a biodegradable polymer)

WHAT IS THE PROBLEM?
Cellulose aggregates easily with itself. It is incompatible with nearly anything other than the lignin/hemicellulose matrix in the plant cell wall.
→ Loss in surface area → No nanocomposite anymore

# Nanocomposite

## Natural Rubber latex membrane for glove applications

| Market | Revenue | Growth (CAGR) |
|---|---|---|
| Medical gloves | $100 M | 9.2% |

Rubber product with good tensile strength and toughness, but very high flexibility

↓ Desired properties

Product that is resistant to breakage but also allows for a close fit

Solution → Reinforce latex with fillers

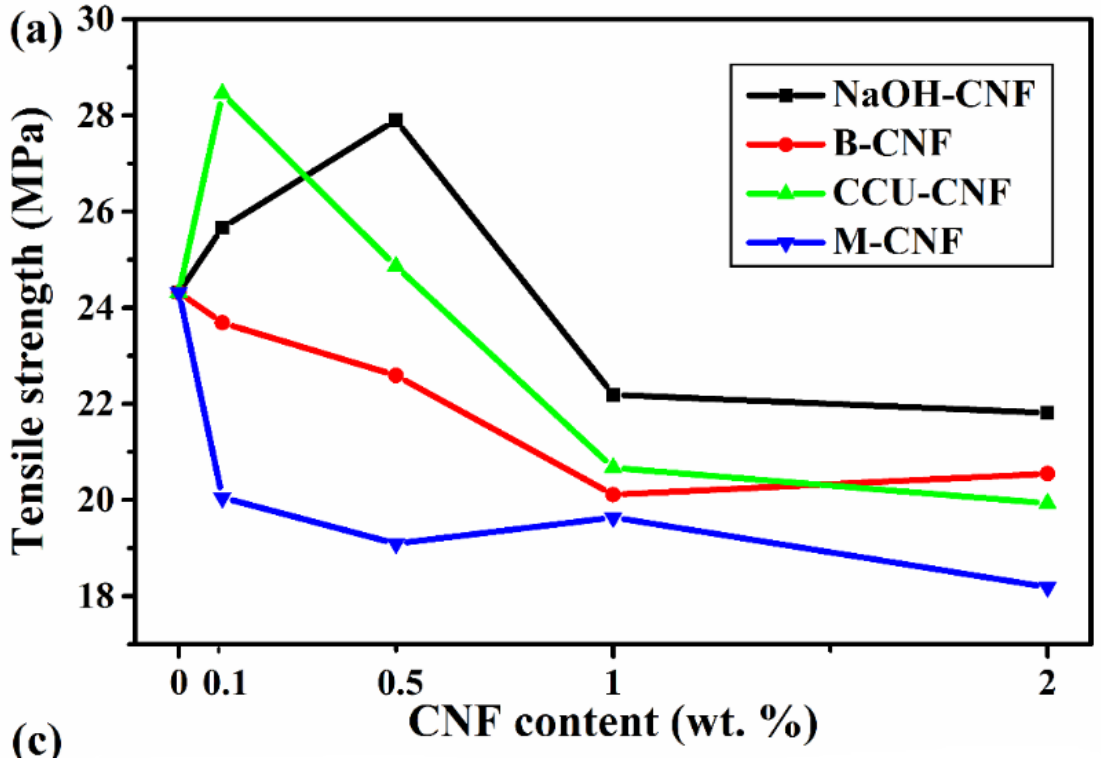
↓ Results

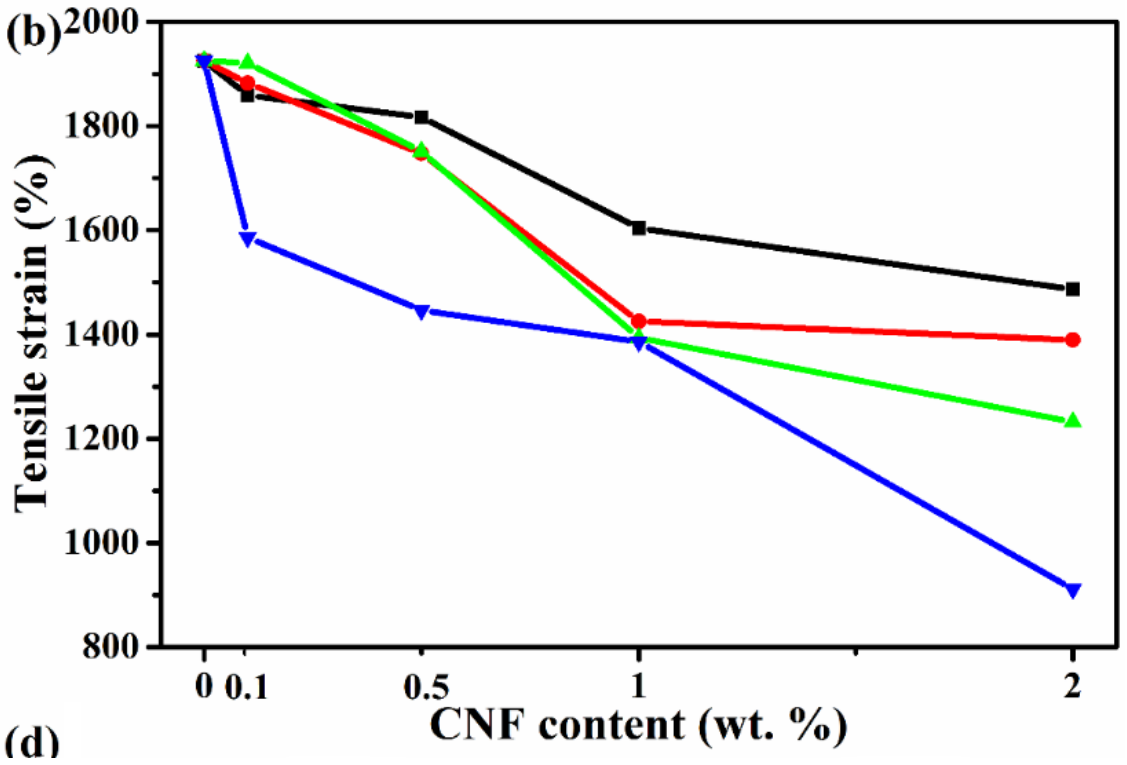
- Increase in tensile strength
- Undesirable increase in stiffness
- Reduction in elongation at break

A solution to this is using SOFT + TOUGH nanofillers = THIN + TACTILE membranes

# Polymer reinforcement

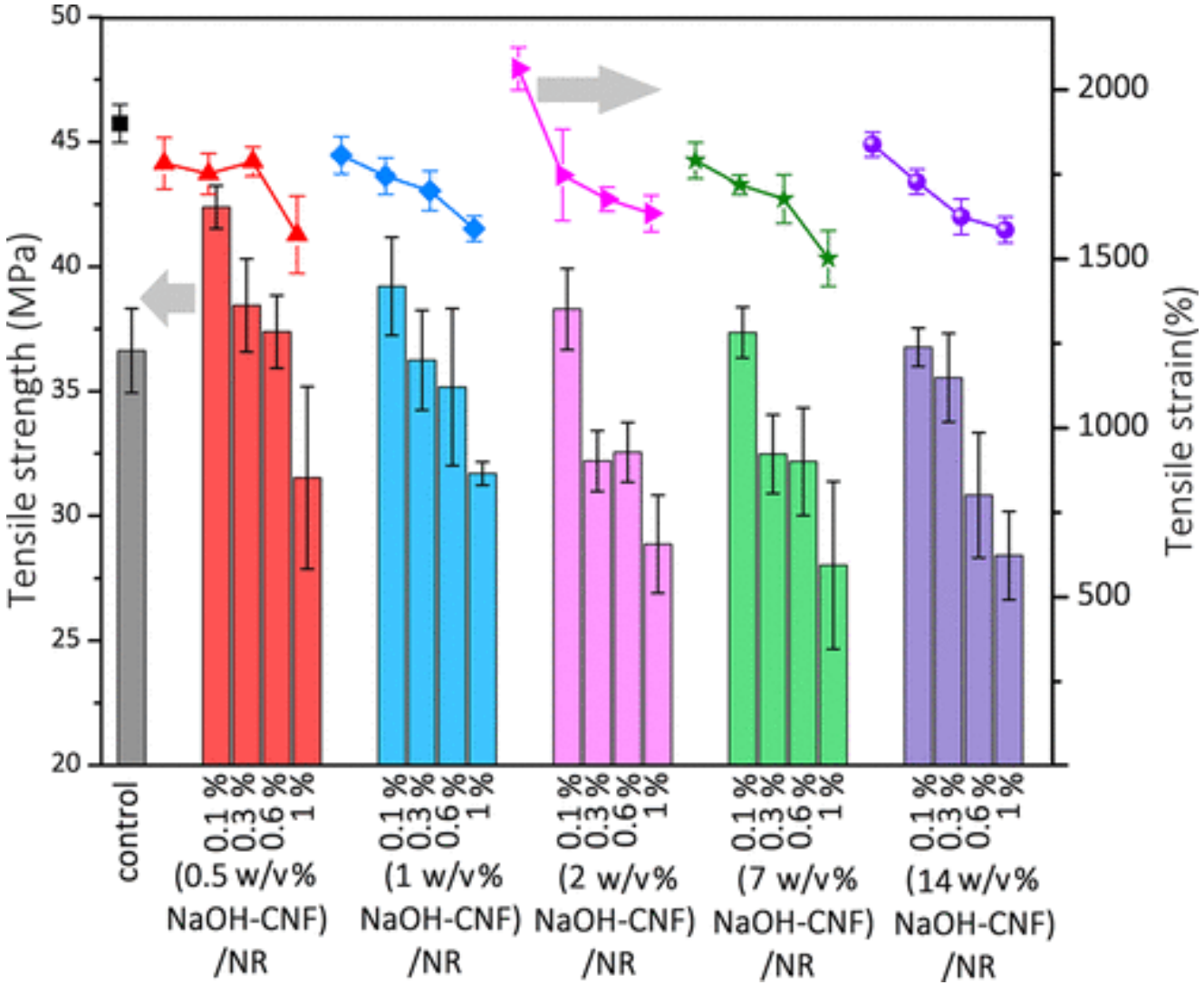
| Sample name | NaOH-CNF | B-CNF | CCU-CNF | M-CNF | Control sample NR |
|---|---|---|---|---|---|
| **Contact angle (θ)** | 45° ± 4.6° | 30° ± 3.5° | 62.7° ± 5.1° | 8.6° ± 1.4° | 82.7° ± 3.1° |

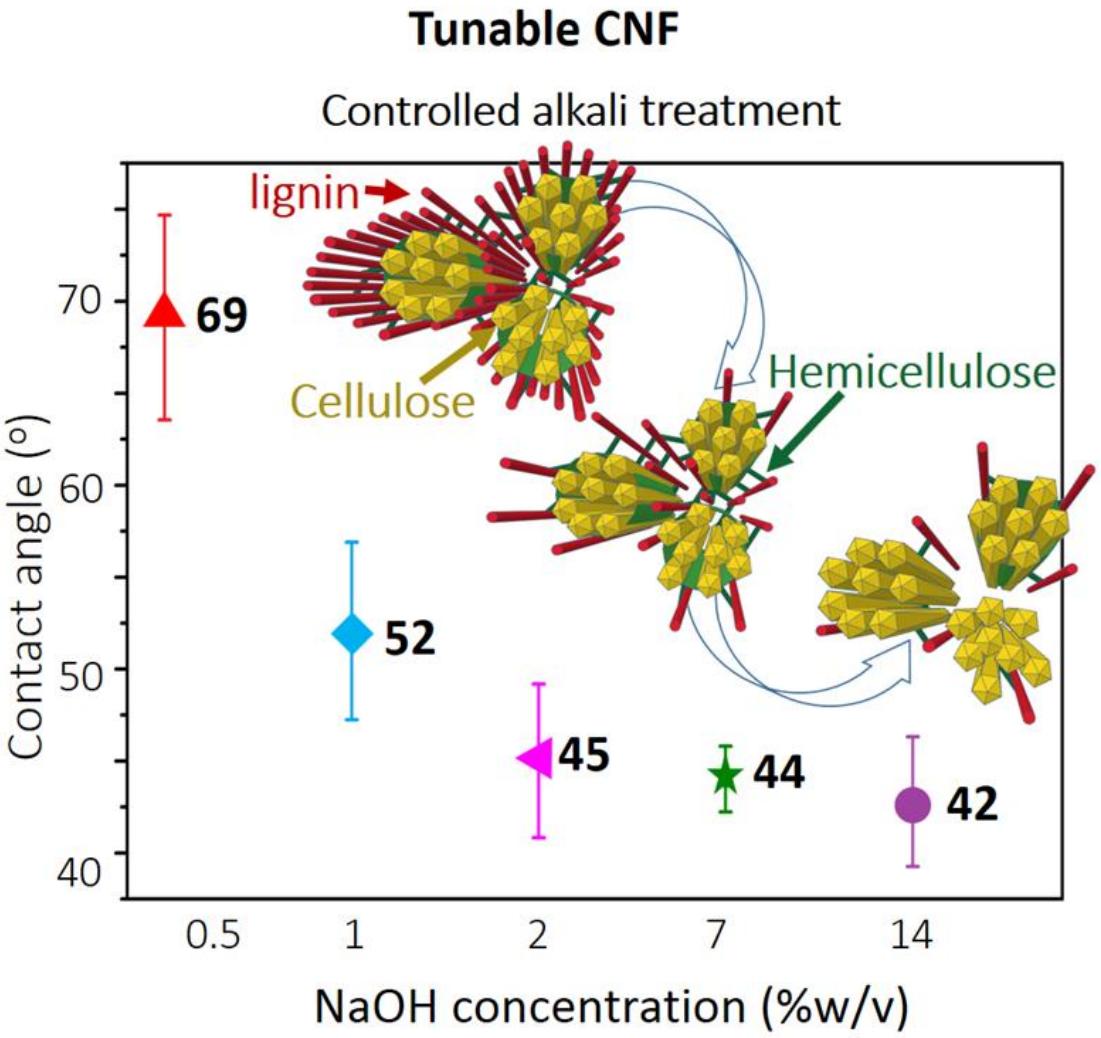



Nanoscale, 2017, 9, 9510–9519

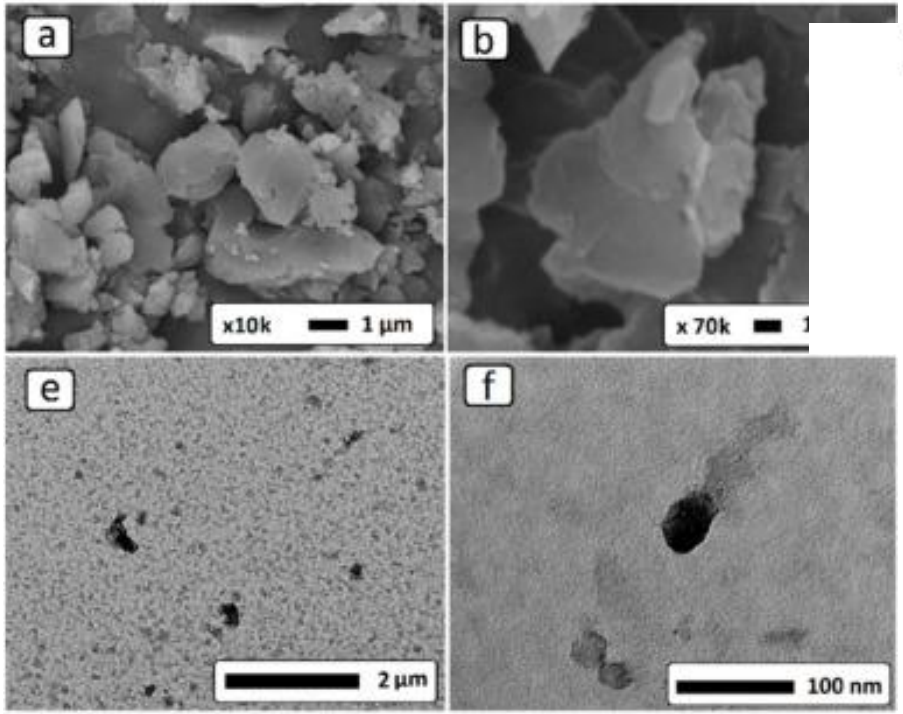
# Polymer reinforcement

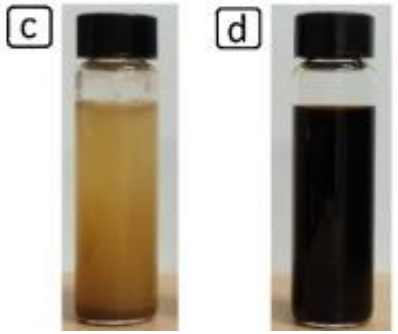


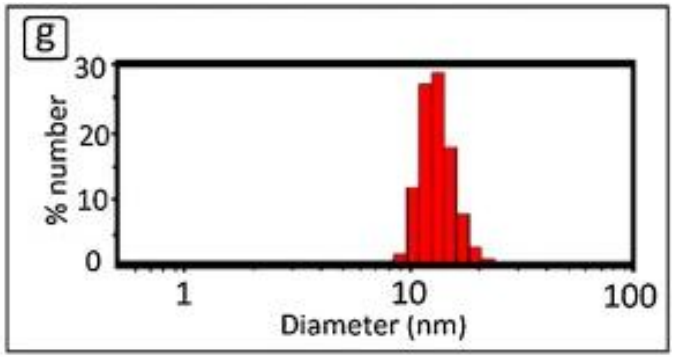



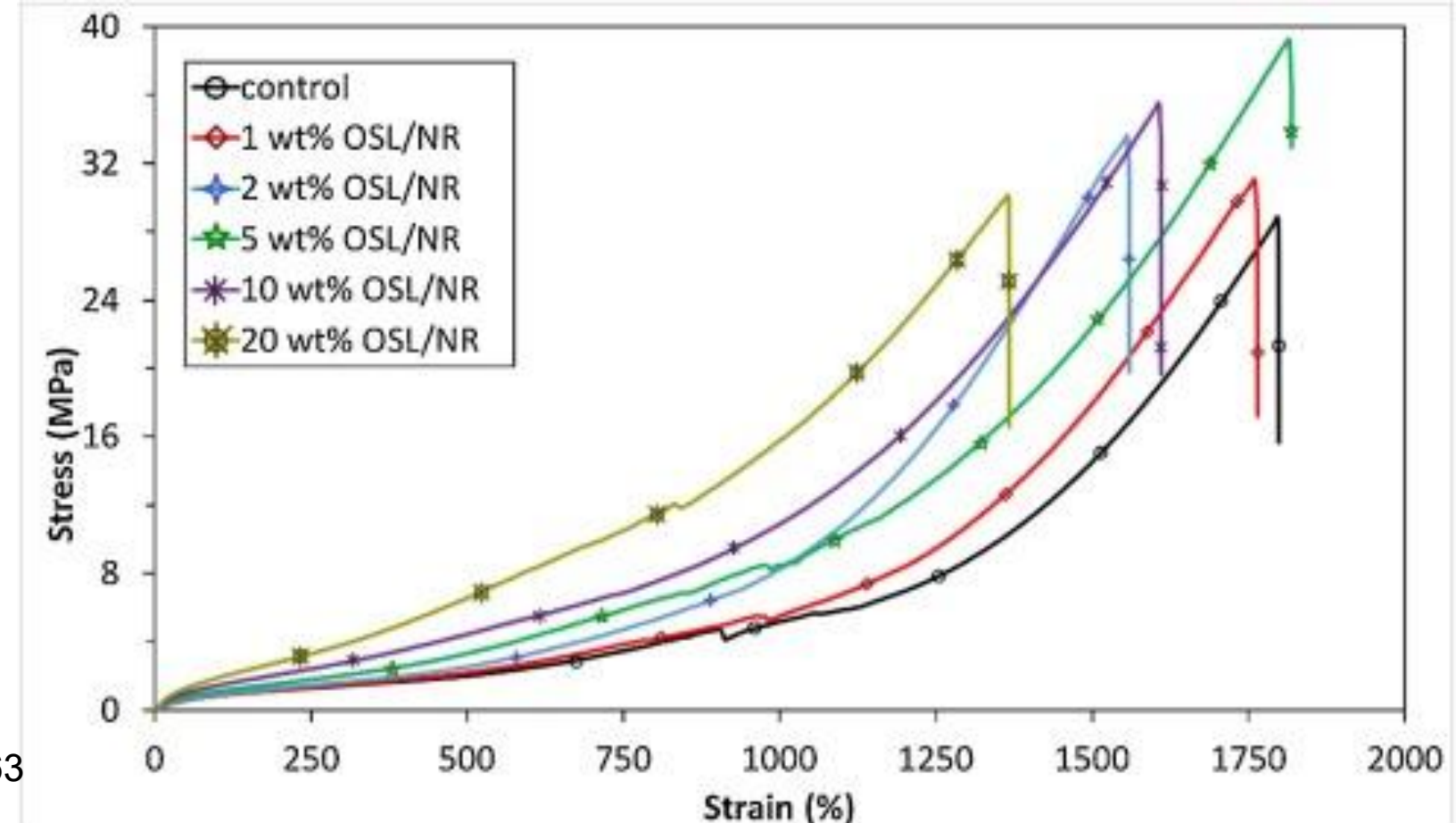

ACS omega 2019 3 (11), 15933-15942

# Polymer reinforcement




Industrial Crops and Products, 2021 159, 113063

# Case study: a self-disinfecting cellulose coating

*Antimicrobial and low-fouling materials*

# Antimicrobial and low-fouling materials

**Surface access**

active groups must be exposed at the interface

**Killing mechanism**

contact-killing, ion release, ROS, low-fouling

**Durability**

adhesion, abrasion, leaching, cleaning cycles

**Safety**

cytotoxicity, biocompatibility, migration, regulation

# Antimicrobial resistance why new materials matter

## The AMR crisis

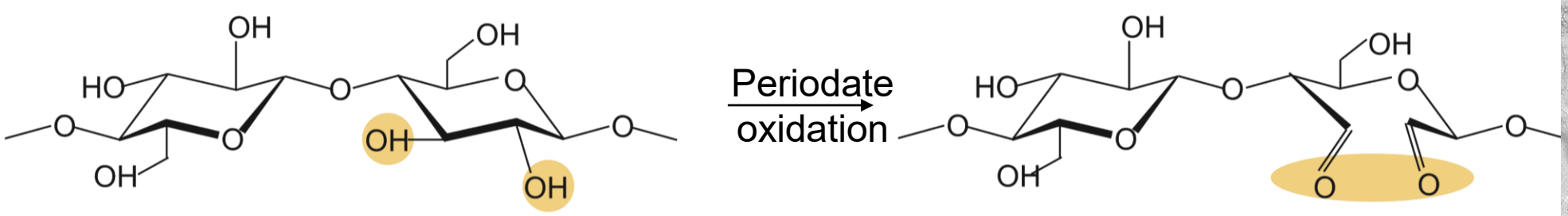
- ~1.3 million deaths directly attributed to AMR infections in 2019 (Lancet 2022) — projected to exceed cancer mortality by 2050.
- Conventional antibiotics: slow pipeline, rapid resistance development.
- Materials-based solutions: antimicrobial surfaces and coatings can kill or inhibit bacteria without relying on antibiotics that drive resistance.
- Key targets: hospital-acquired infections, food packaging (extending shelf life without preservatives), wound dressings.

## Why nanocellulose?

- Surface OH groups → easy, tuneable chemistry to attach antimicrobial agents.
- High surface area → more contact surface per gram than conventional materials.
- Biocompatible → safe for wound contact, food contact.
- Biodegradable → avoids the long-term persistence problem of plastic antimicrobial films.
- Can be formed into films, hydrogels, aerogels, coatings - multiple delivery formats.

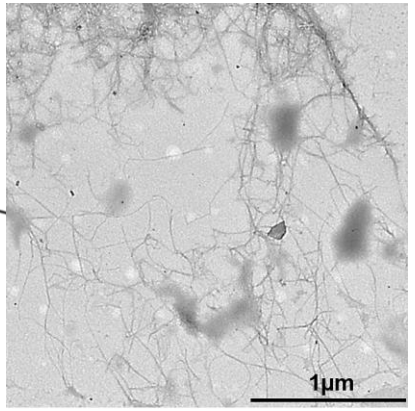
# Our platform solutions

## Alteration of cellulose backbone to antimicrobial functionalities

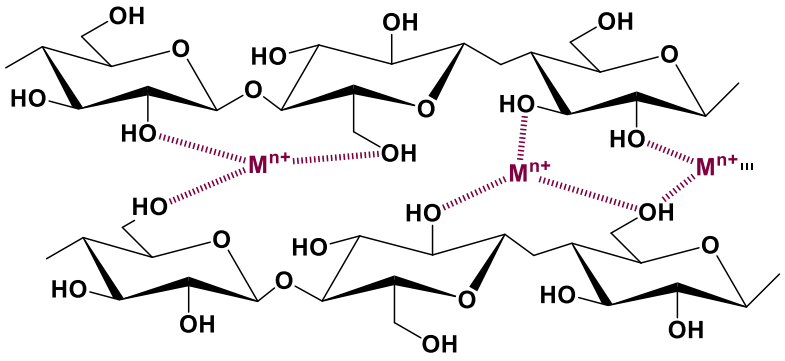


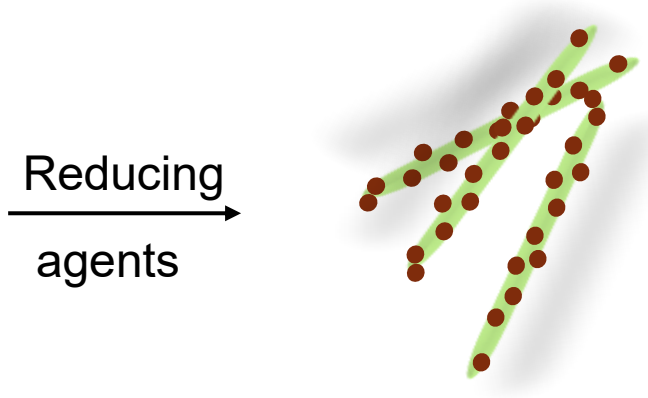

Cellulose

Dialdehyde cellulose

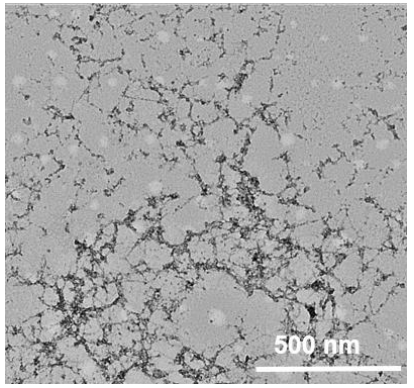



## Metal nanoparticles templating on nanocellulose surface





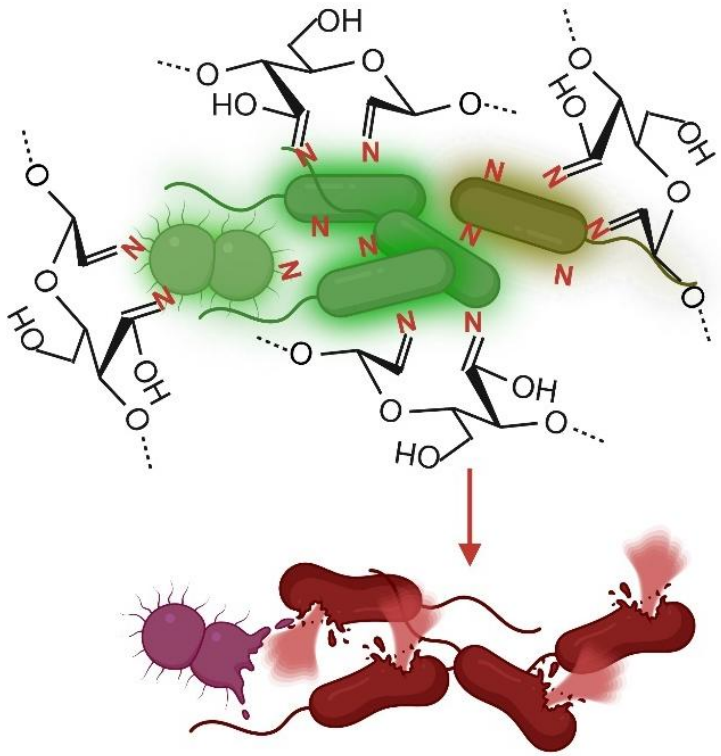

Metal NPs on Nanocellulose



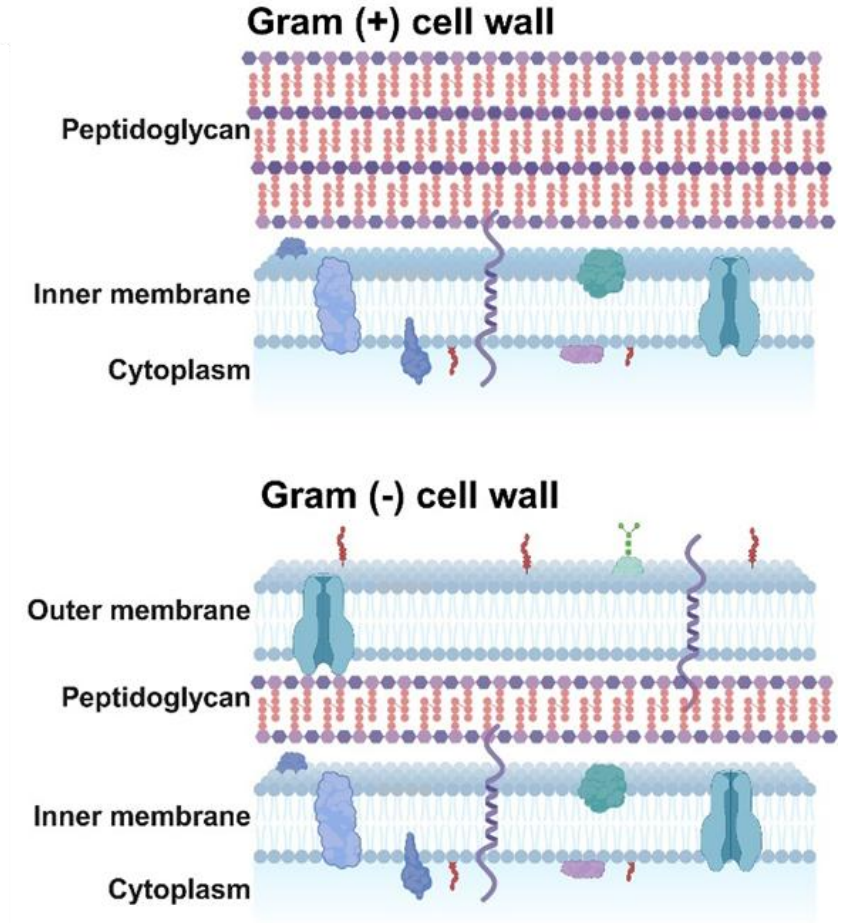

*ACS Appl. Mater. Interfaces,* 2025, 17 (10), 16027-16039

# Antibacterial mechanism of dialdehyde cellulose

**Antibacterial mechanism of dialdehyde cellulose**



**Structural comparison between Gram (+) & Gram (-) bacterial cell walls**




*ACS Appl. Mater. Interfaces,* 2025, 17 (10), 16027-16039

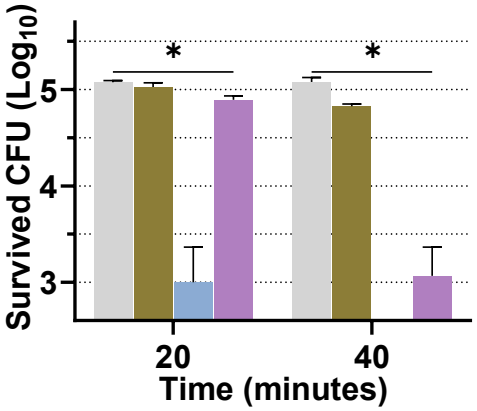
(created with Biorender.com)

# Antibacterial efficacy of dialdehyde cellulose

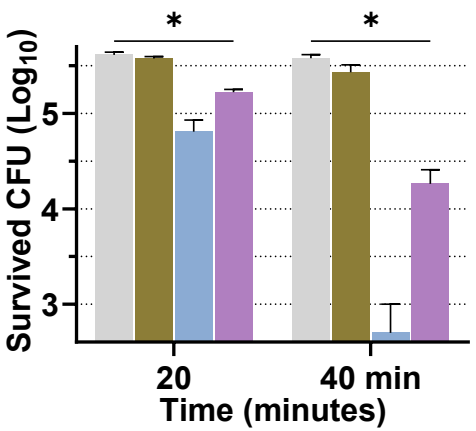


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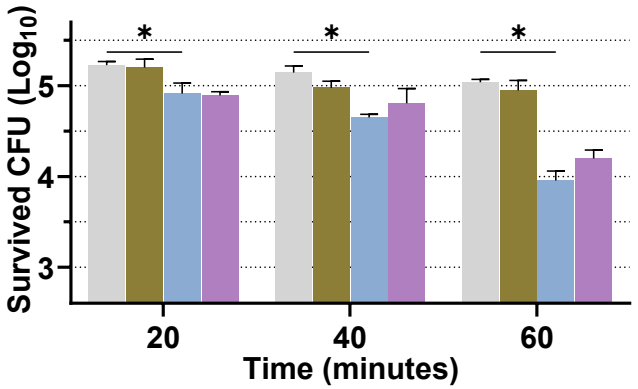
**S. Aureus** (Gram +)



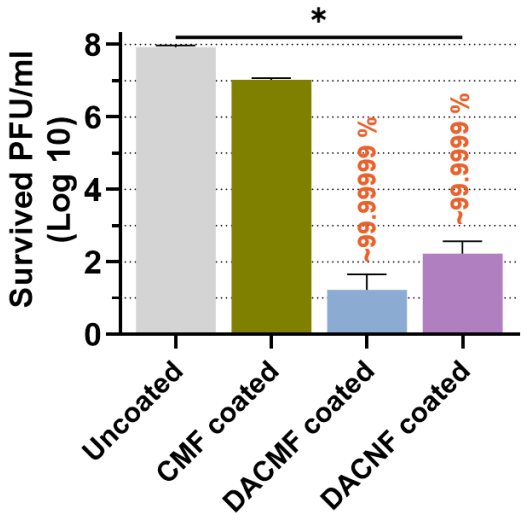

**MRSA** (Gram +)




**E. Coli** (Gram -)

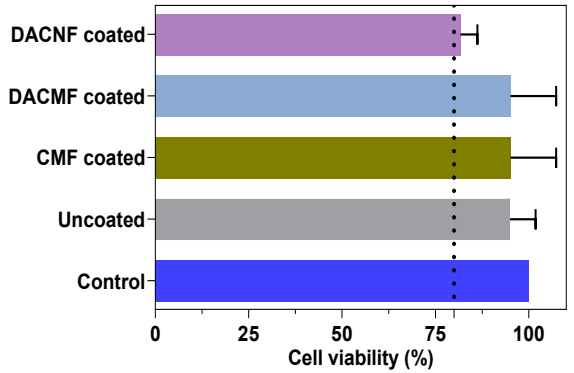



**A/H1N1 (2009)** influenza virus




Uncoated CMF DACMF DACNF

**Cytotoxicity**

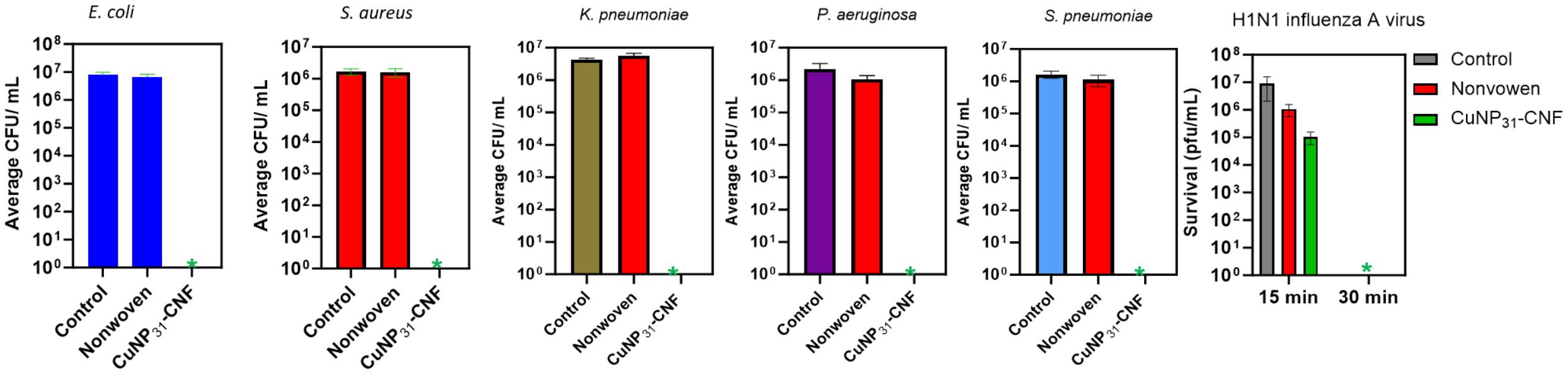


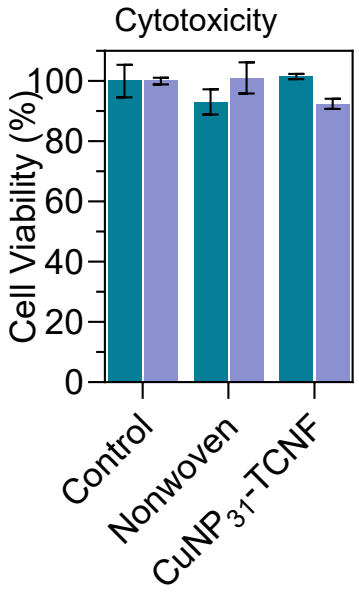

ISO 10993-5, using L929 cell line- mouse fibroblast

*ACS Appl. Mater. Interfaces,* 2025, 17 (10), 16027-16039

# Antibacterial properties of copper oxide nanocellulose






ISO 10993-5, using L929 cell line- mouse fibroblast

Nanoscale 18 (35), 19165-19177

# What changes when you leave the lab?

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**Lab scale (mg–g)**

Prove the concept. Optimise chemistry. Characterise the material. Cost and yield don't matter yet. This is where most academic research lives.

**Pilot scale (kg–100 kg)**

Prove the process. Batch consistency becomes critical. Energy consumption is now measurable and expensive. This is where most promising materials hit their first wall.

**Commercial scale (tonnes)**

Prove the business. Feedstock supply chain, regulatory compliance, customer qualification, and unit economics all must work simultaneously.

As you move from the laboratory to manufacturing, the dominant questions shift from material performance to process reproducibility, cost, supply reliability and product qualification.

# Scale-up bottlenecks for biomass-derived materials

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**Feedstock variability**

season, geography, moisture, ash, contaminants

**Pretreatment intensity**

chemicals, water, energy, waste streams

**Yield and concentration**

low solids increase downstream cost

**Product reproducibility**

batch-to-batch quality and specifications

**Drying and redispersion**

often hidden cost and performance bottlenecks

**Customer adoption**

incumbent materials are cheap, optimised and qualified

# Manufacturing readiness: what must be controlled?

**Input control**

feedstock specification, supplier, storage, contamination

**Process control**

unit operations, critical parameters, mass balance

**Quality control**

identity, purity, functional performance, stability

**Risk control**

toxicity, migration, exposure, worker safety, waste streams

**Documentation**

traceability, SOPs, change control, claims evidence

# From manufacturing readiness to responsible manufacturing

Scaling a biomass-derived product is not only a technical challenge. Governance, ownership and benefit-sharing also shape whether a value chain can translate responsibly.

| Technical readiness | Partnership readiness | Translation readiness |
| --- | --- | --- |
| Can the feedstock, process and product be controlled reproducibly at scale? | Who has access to the resource, who helps define value, and how are decisions made? | Can the pathway scale in a way that is ethical, acceptable and beneficial to all parties? |

**A technically scalable process is not automatically a responsible biomanufacturing system.**

**Question to keep in mind**

If a biomass resource is connected to Indigenous knowledge, what must be agreed before the technology is developed further?

# Biomass manufacturing opportunities for Australia

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### Local feedstocks

Sugarcane residues, forestry waste, cotton and grain waste, algae and seaweed

### High-value products

Nanocellulose, biopolymers, functional materials, biofuel, composites and specialty chemicals

### Regional opportunity

Manufacturing close to biomass sources, skilled jobs and new revenue for producers

## Where Australia can compete

- Build on local biomass rather than imported feedstocks: sugarcane bagasse and trash, forestry residues, cotton and grain waste, algae and seaweed.
- Target high-value markets where Australia has research and industry strengths: sustainable packaging, biomedical materials, composites, mining sorbents and specialty chemicals.
- Close the scale-up gap through pilot manufacturing, feedstock standardisation, product qualification, techno-economic and life-cycle assessment, and regulatory pathways.
- Create regional and Indigenous-led enterprises through equitable partnerships, consent, benefit-sharing and protection of Indigenous knowledge.

CSIRO; Queensland Government biofutures resources; National Reconstruction Fund priority areas

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# Thank you

A/Prof Nasim Amiralian | Group Leader
Australian Institute for Bioengineering and Nanotechnology
n.amiralian@uq.edu.au
07 3443 1296

https://researchers.uq.edu.au/researcher/11106

@AmiralianLab

CRICOS code 00025B

